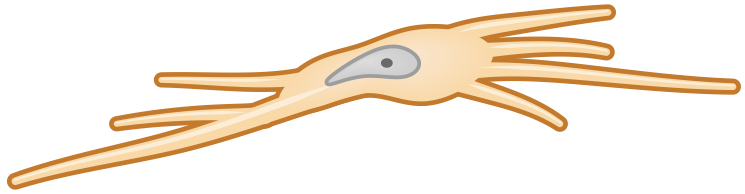
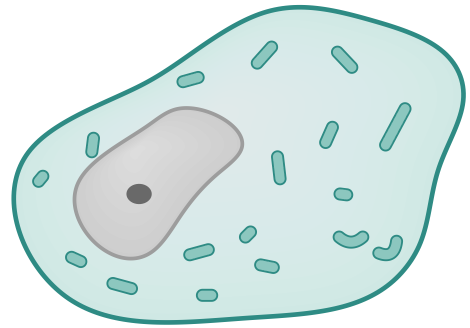
# On the precision of dynamic gene regulation

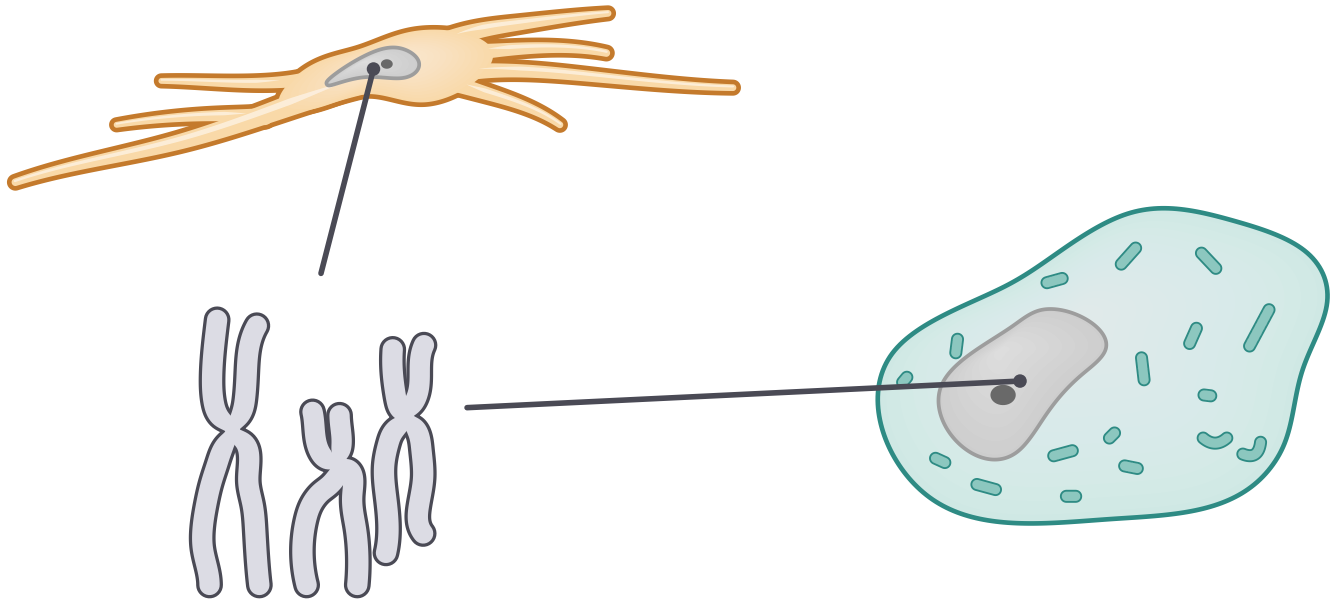
Tarek Tohme

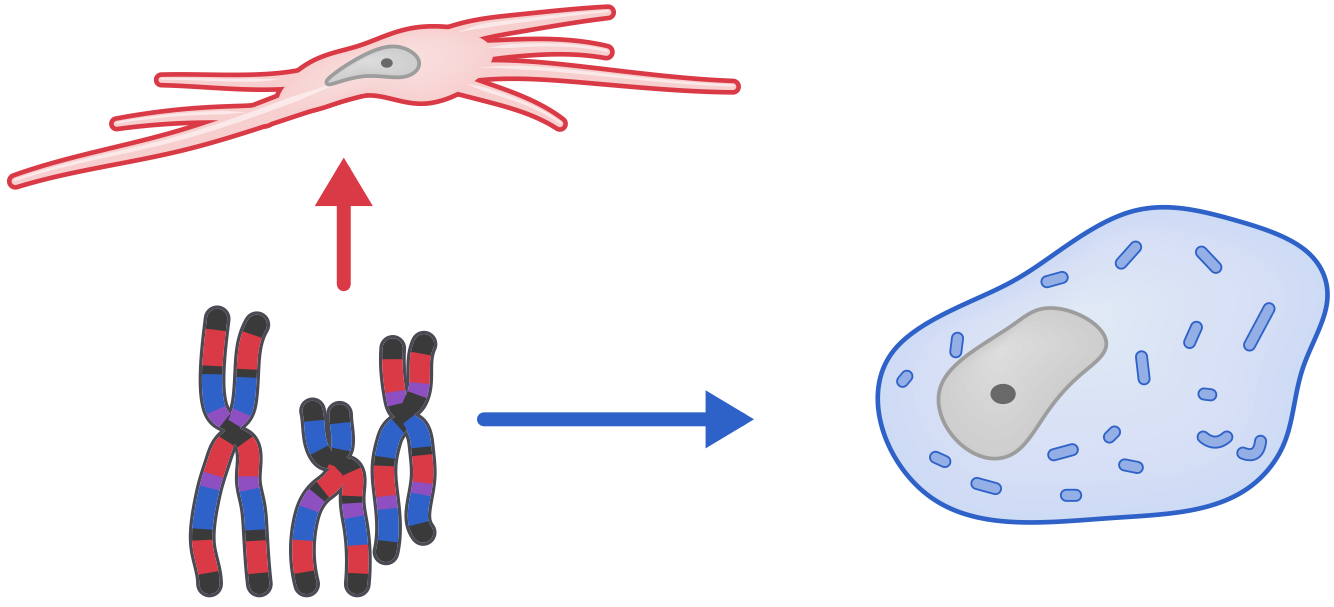
November 3, 2026

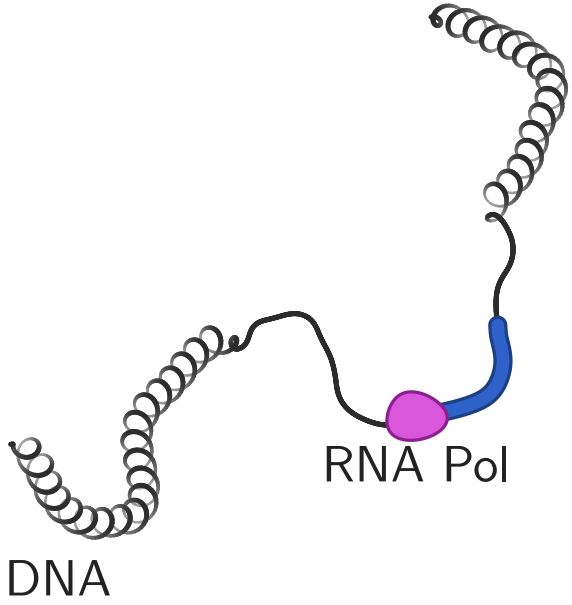




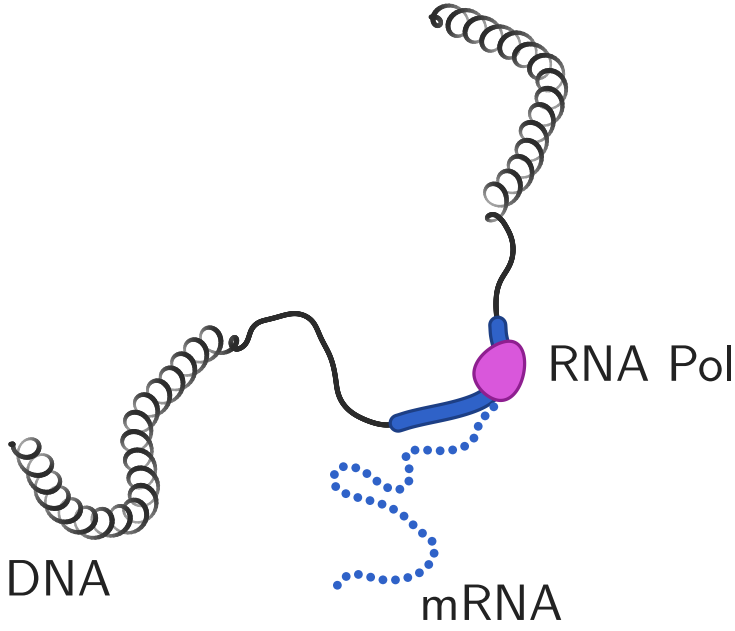




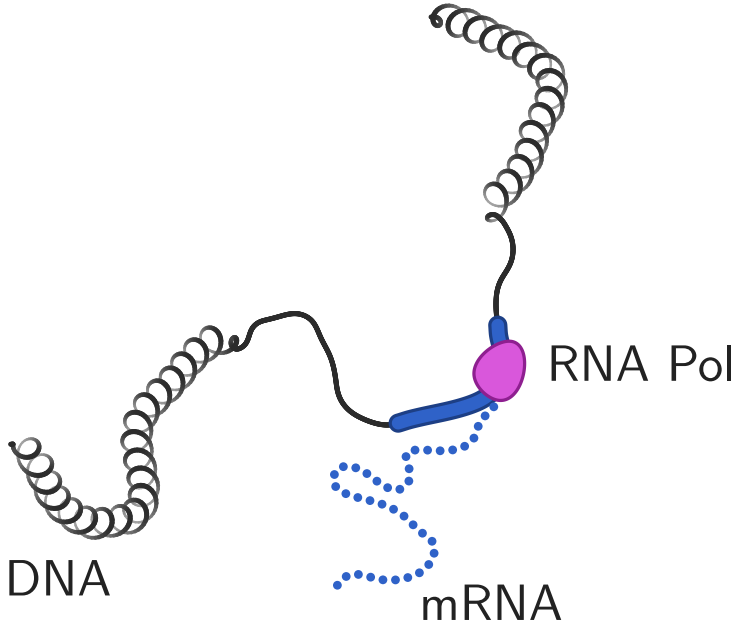




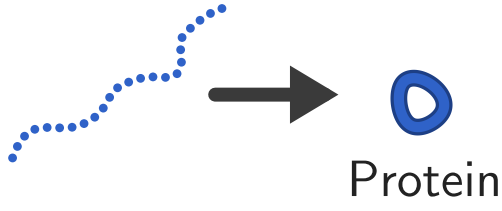

**Transcription**



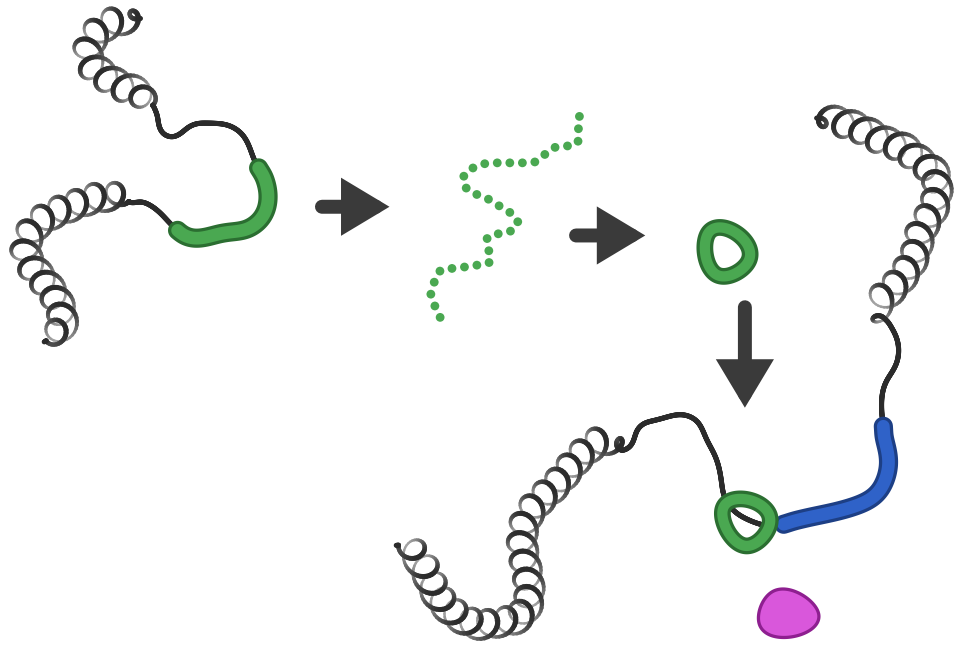

**Transcription**

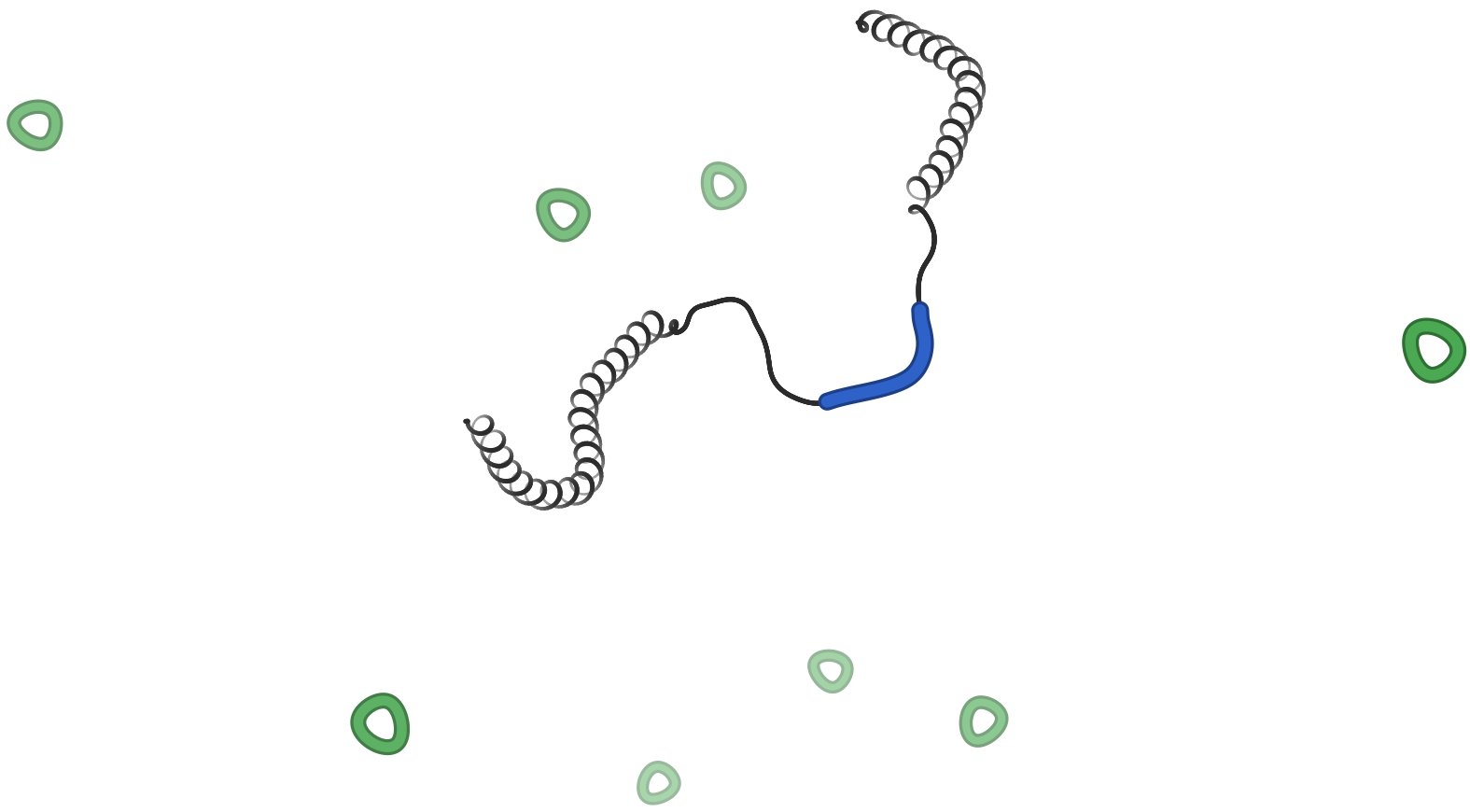


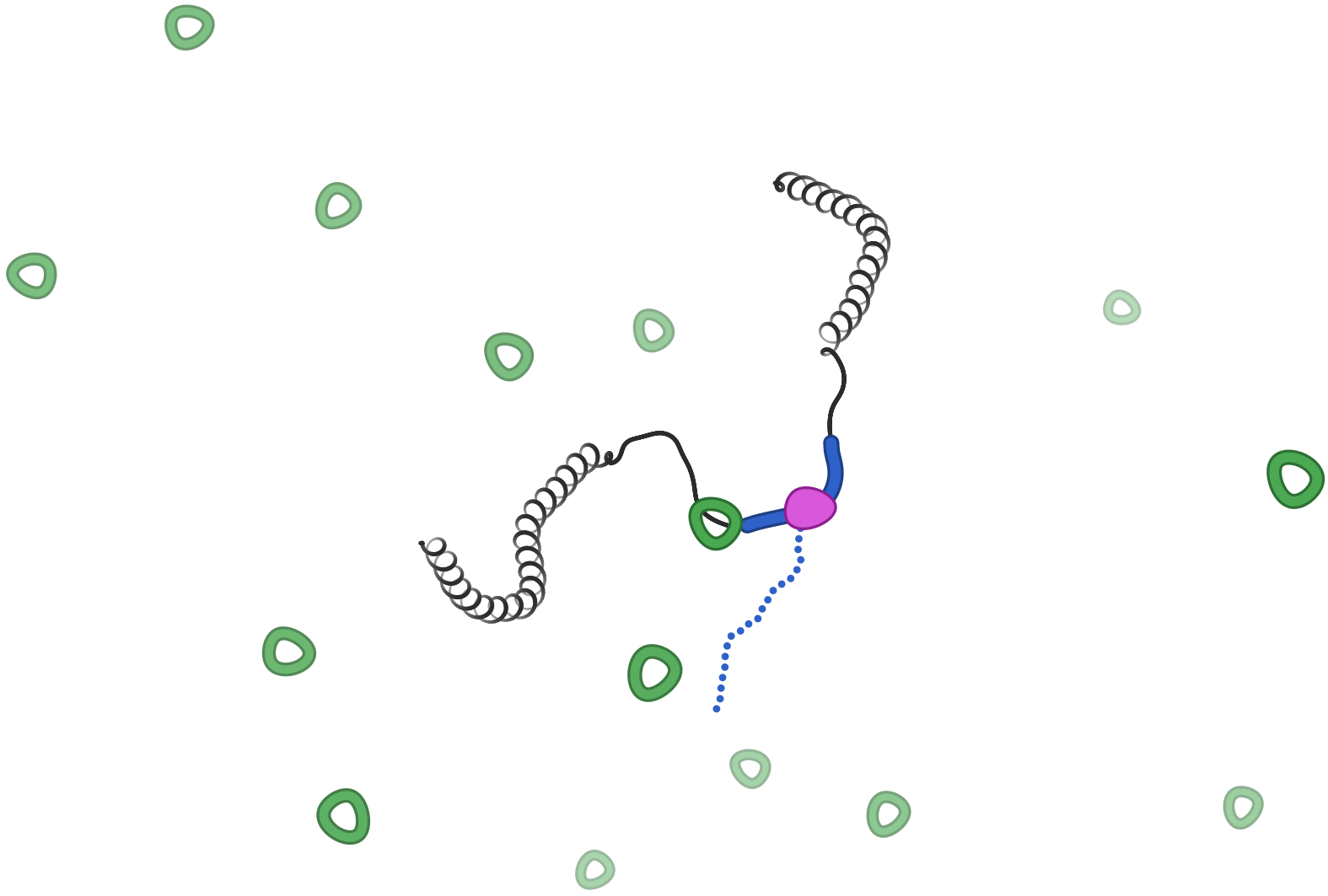

**Transcription**

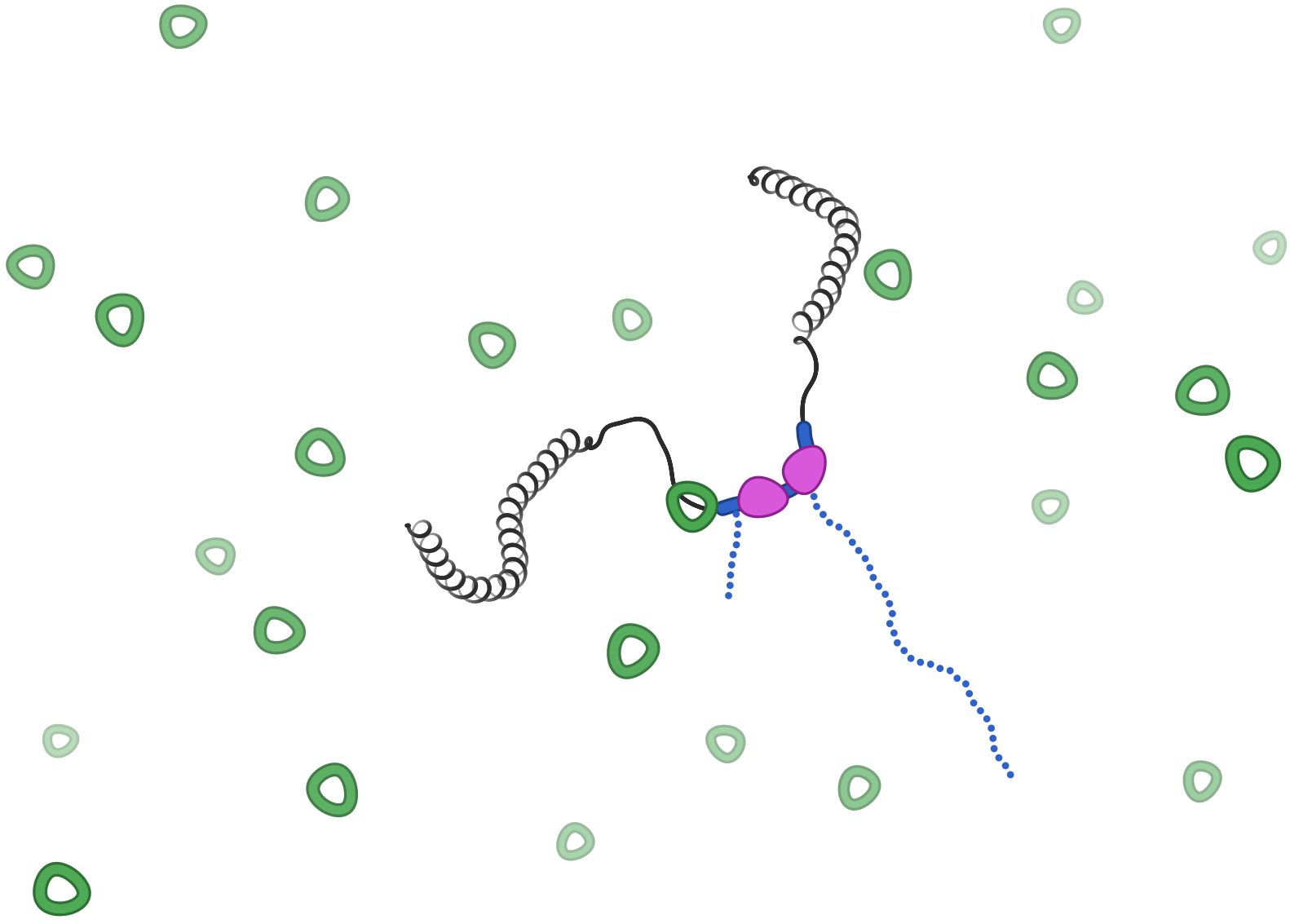


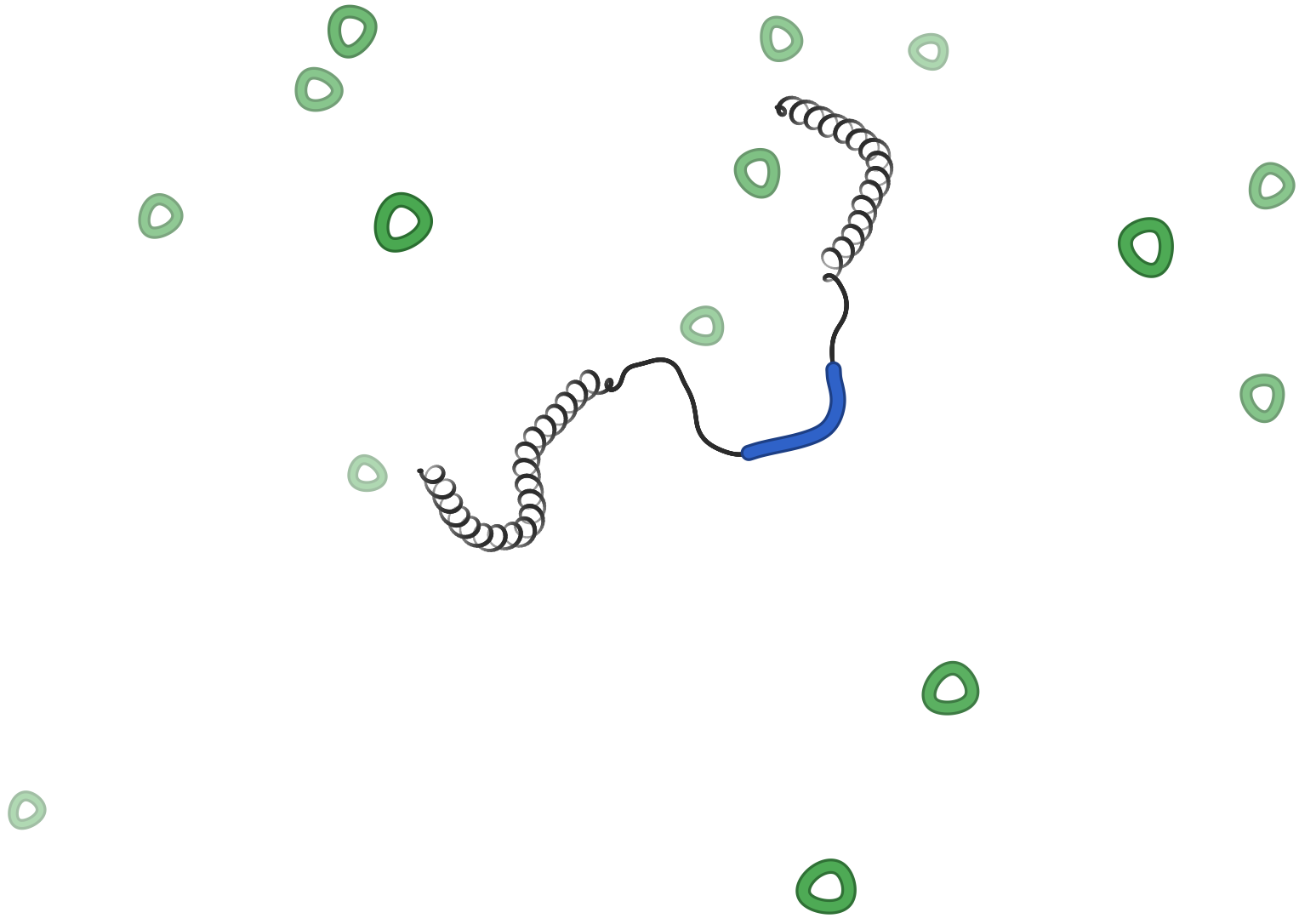

**Translation**

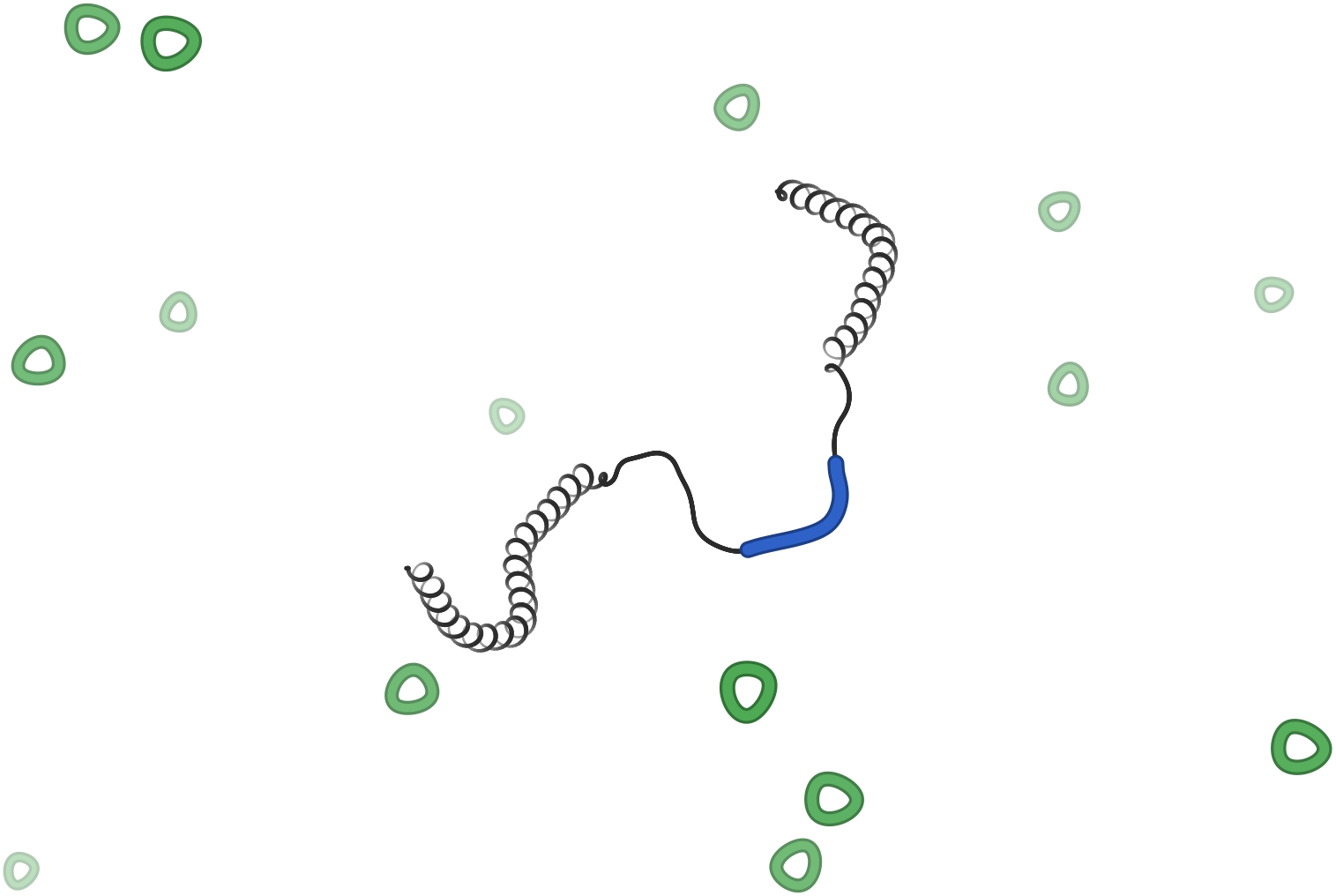


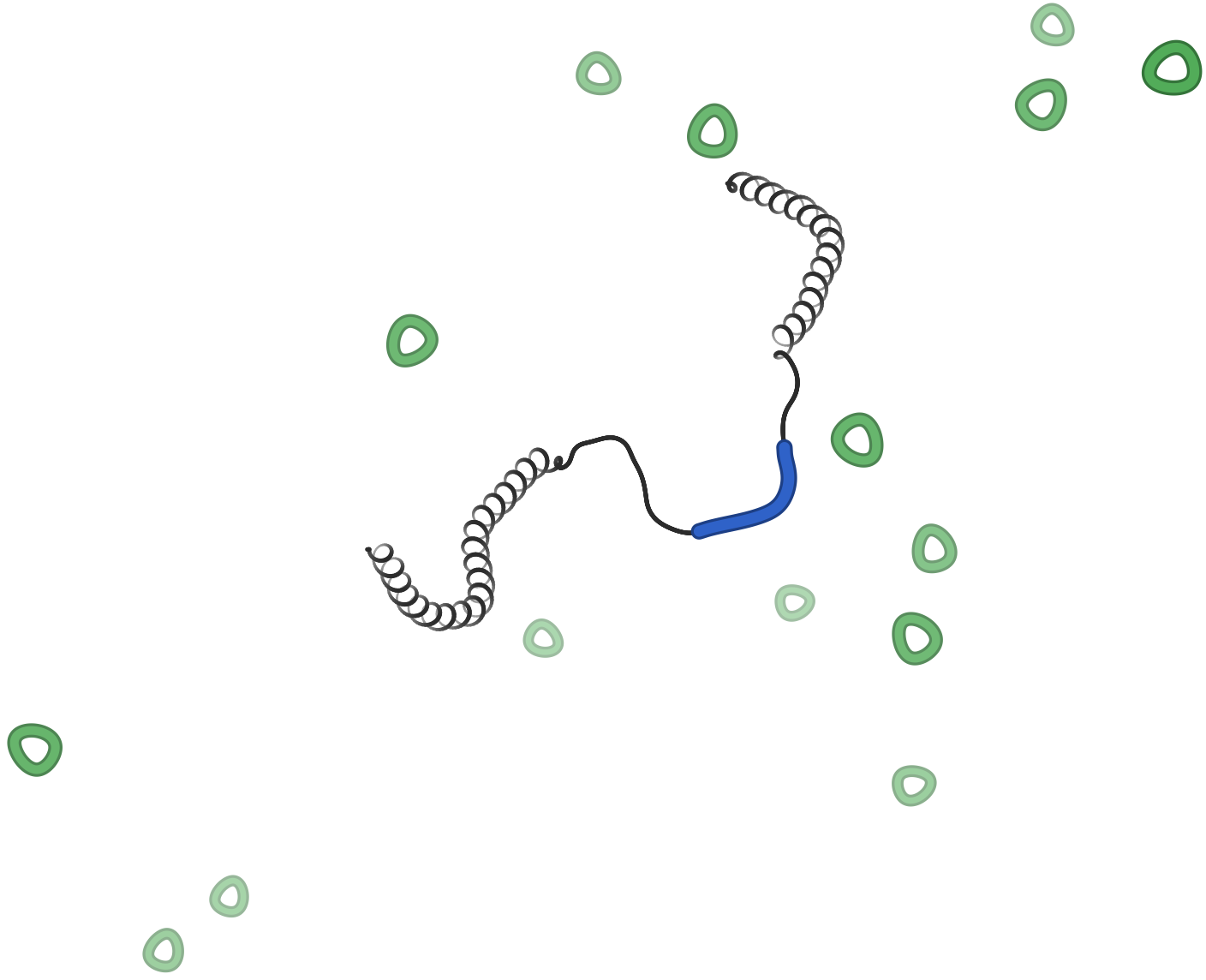


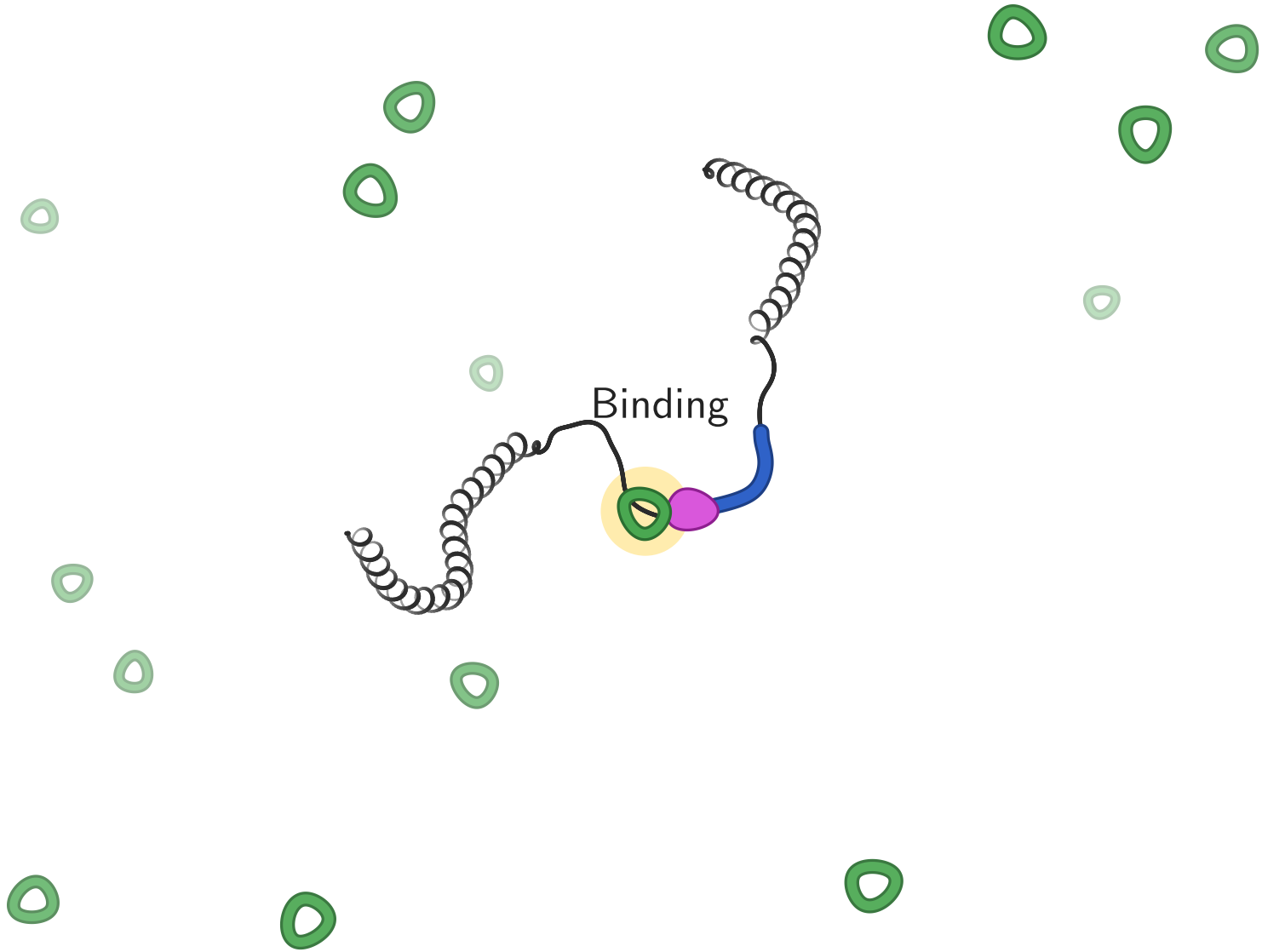


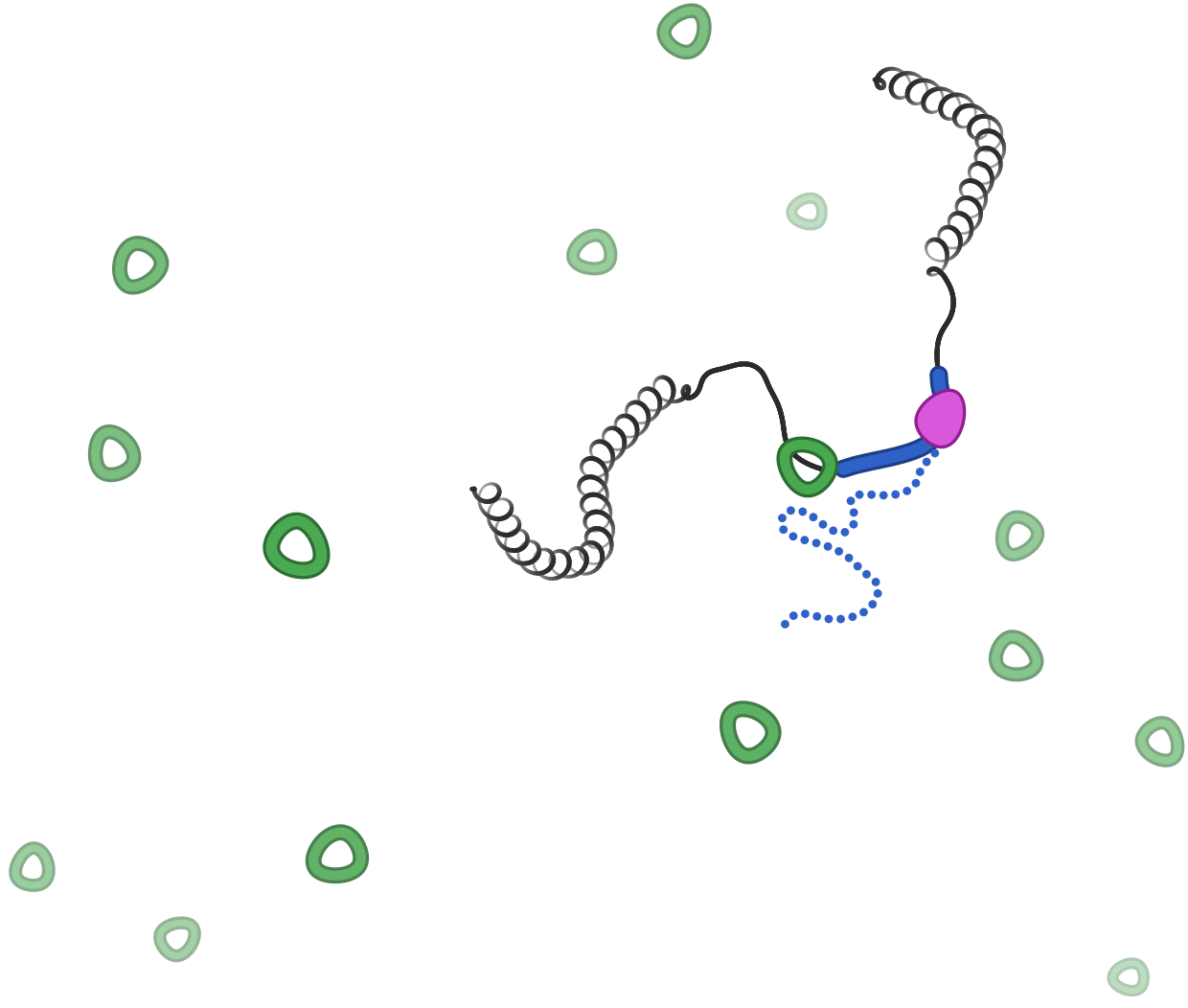


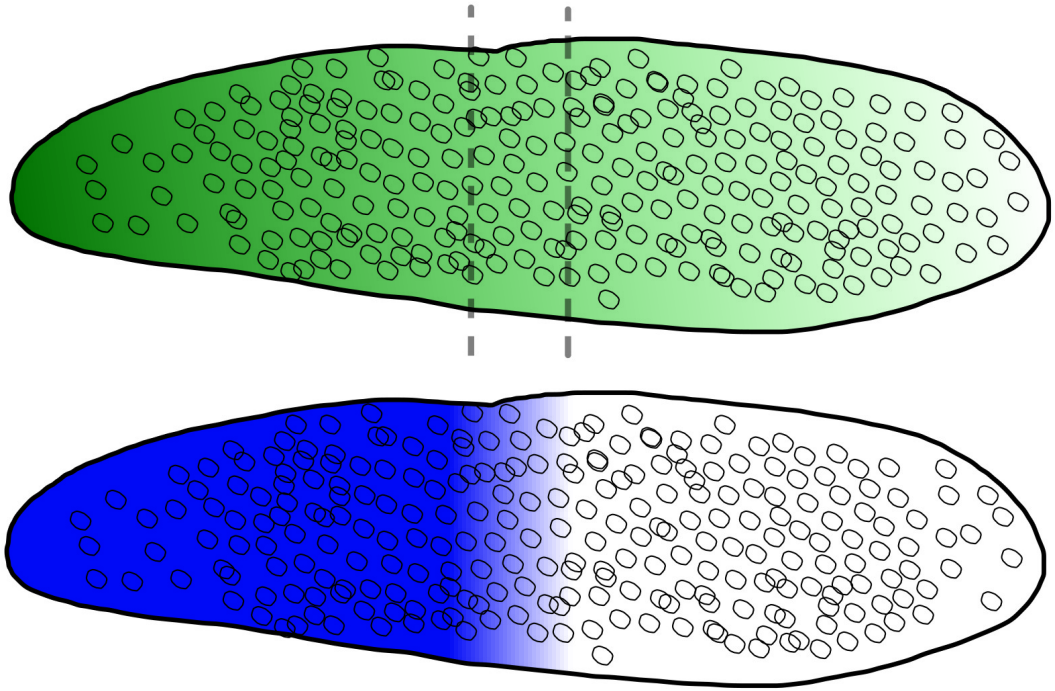






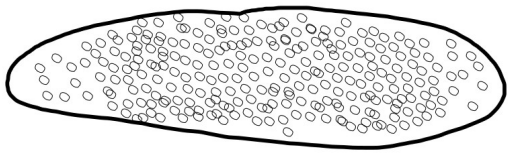
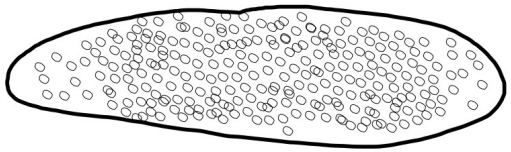

Binding



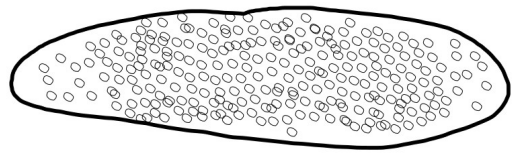
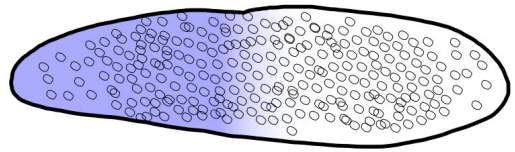


TF concentration

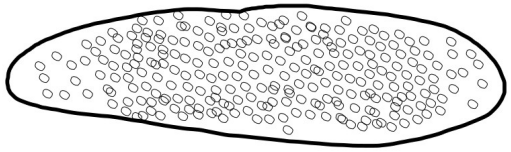
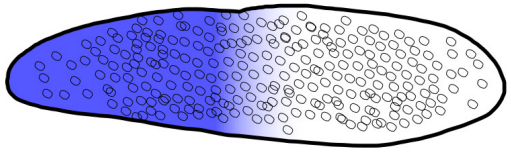
Level of expression
of controlled gene

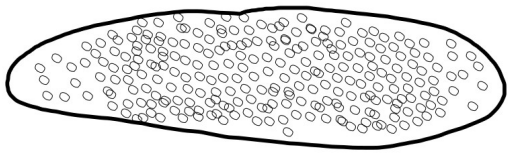
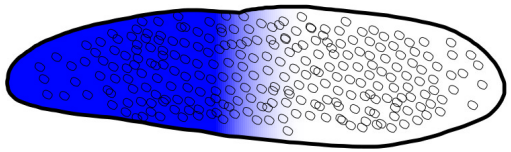
| Division cycle | Embryo |
|---|---|
| **1** |  |
| **2** |  |
| **3** |  |
| **...** | **...** |
| **13** |  |
| **14** |  |

cycle duration:
**4 minutes**

| Division cycle | Embryo |
|---|---|
| **1** |  |
| **2** |  |
| **3** |  |
| **...** | **...** |
| **13** |  |
| **14** |  |

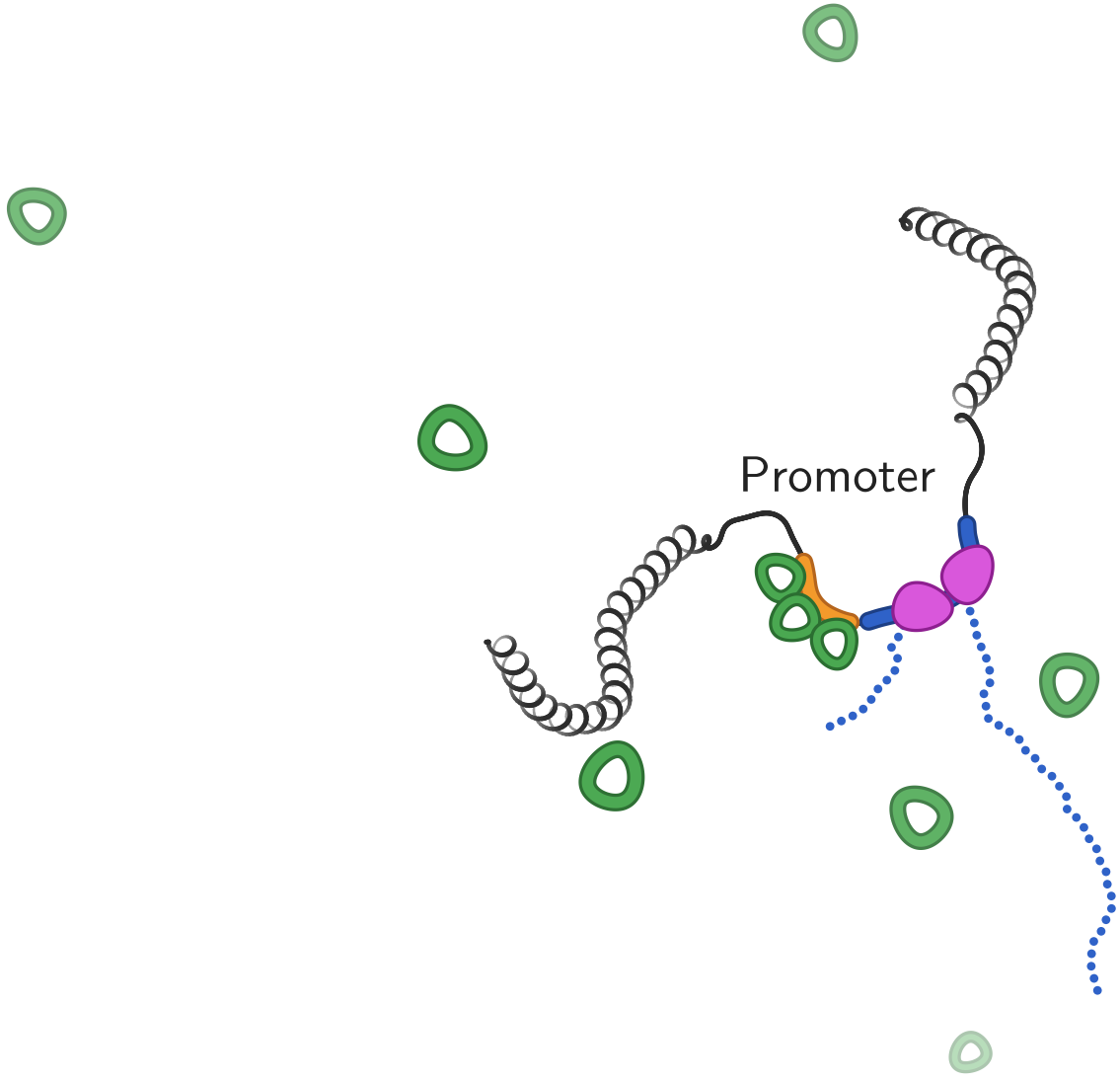
cycle duration:
**4 minutes**

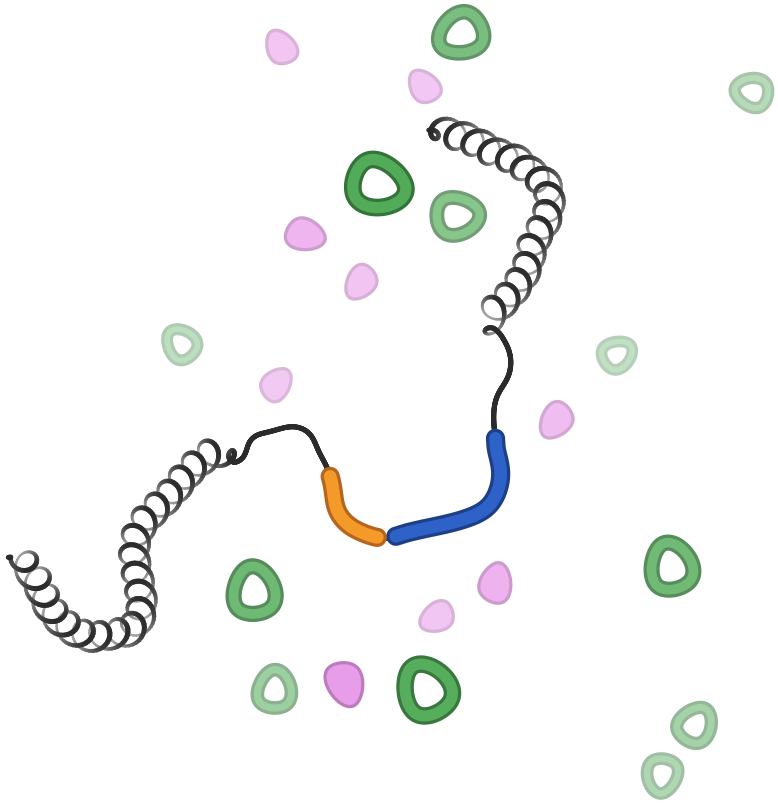
| Division cycle | Embryo | |
|---|---|---|
| **1** |  | |
| **2** |  | |
| **3** |  | |
| **...** | **...** | |
| **13** |  | |
| **14** |  | cycle duration: **4 minutes** |

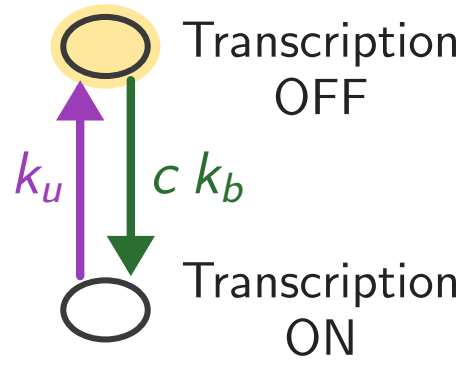
| Division cycle | Embryo |
|---|---|
| **1** |  |
| **2** |  |
| **3** |  |
| **...** | **...** |
| **13** |  |
| **14** |  |

cycle duration:
**4 minutes**

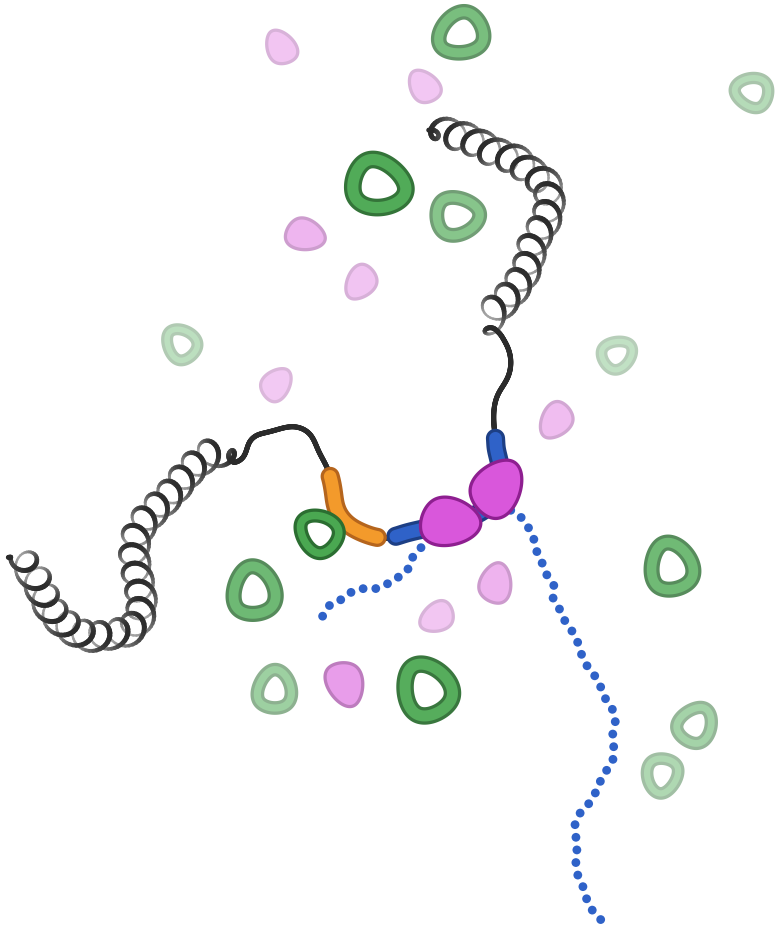


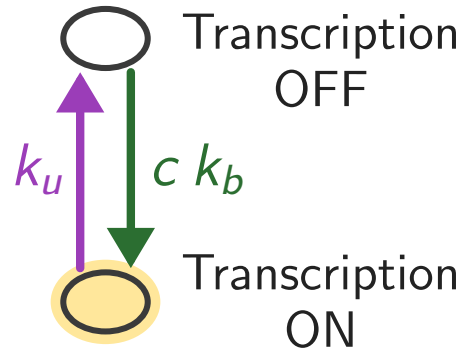

Promoter

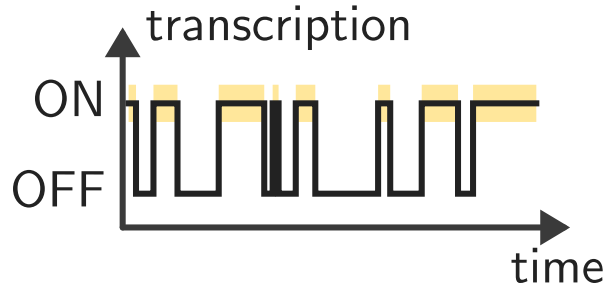


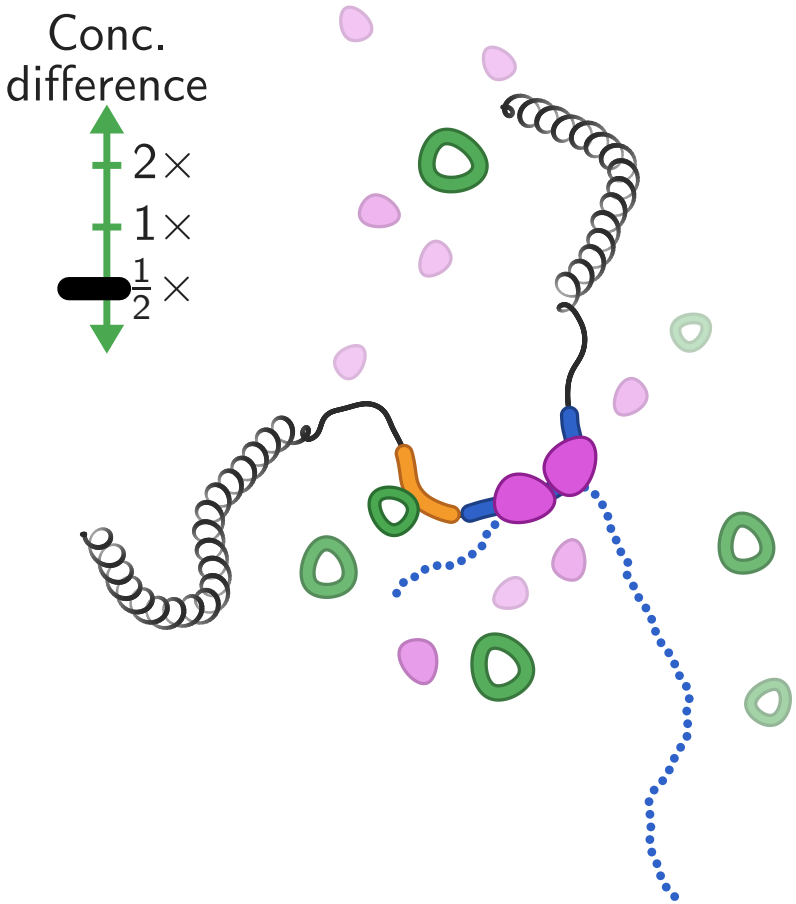

Transcription
OFF
$k_u$
$c\ k_b$
Transcription
ON

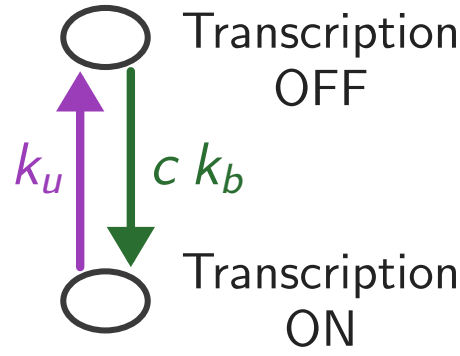

transcription
ON
OFF
time

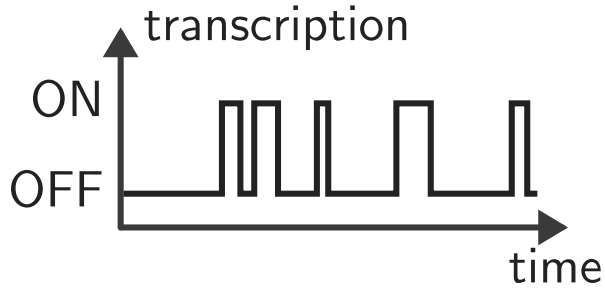


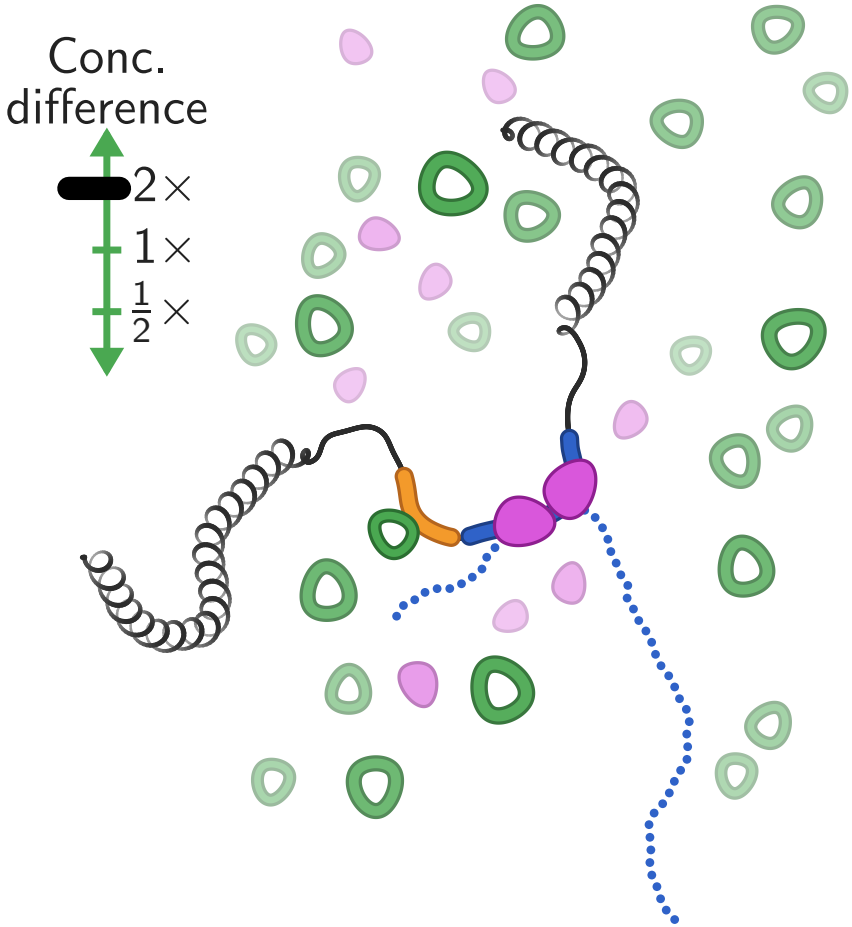

Transcription
OFF
$k_u$
$c\ k_b$
Transcription
ON

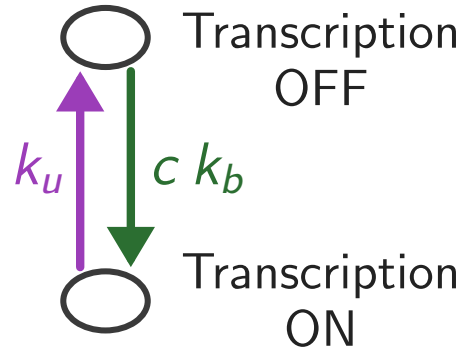

transcription
ON
OFF
time

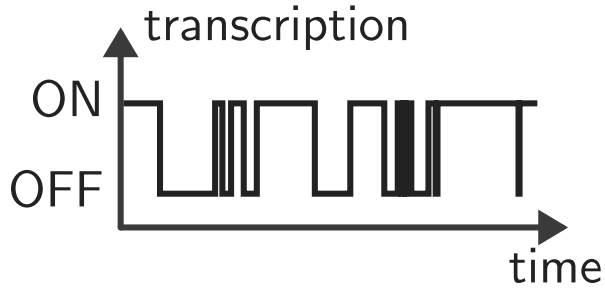

Conc.
difference
2×
1×
1/2 ×

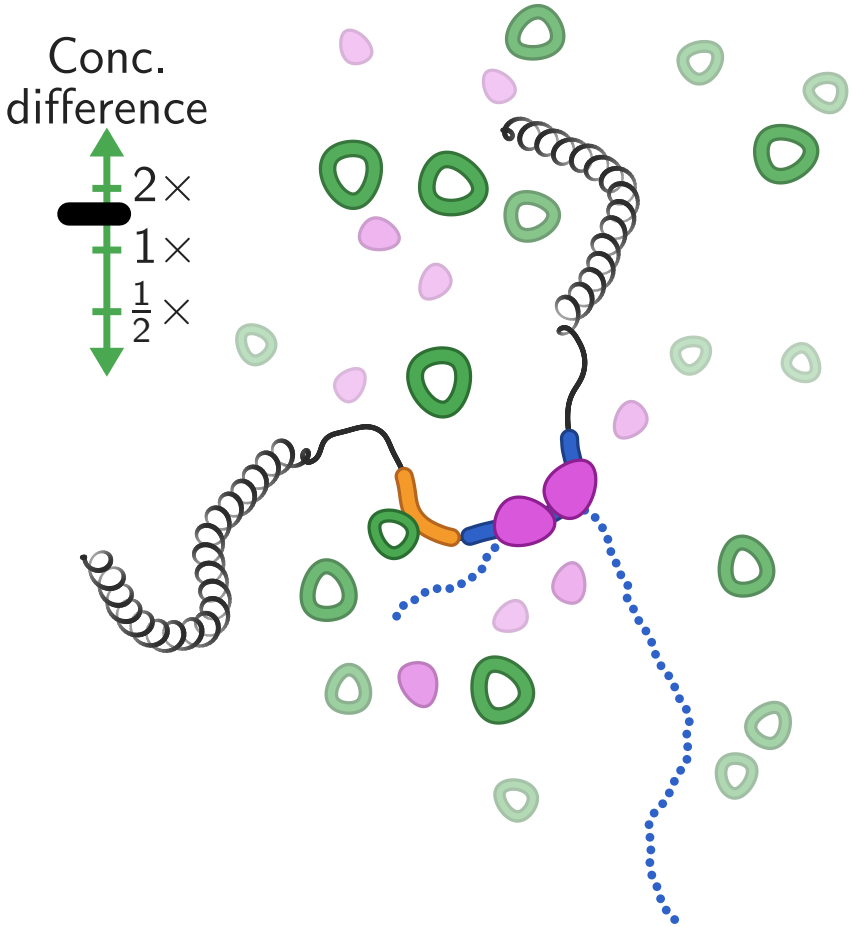

Transcription
OFF
$k_u$
$c\ k_b$
Transcription
ON

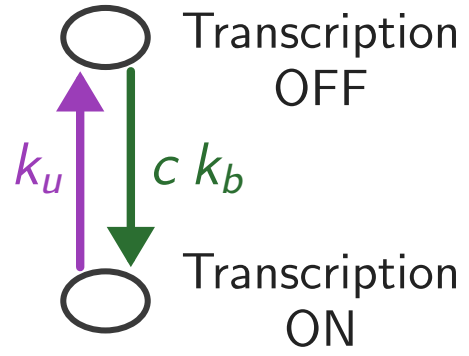

transcription
ON
OFF
time

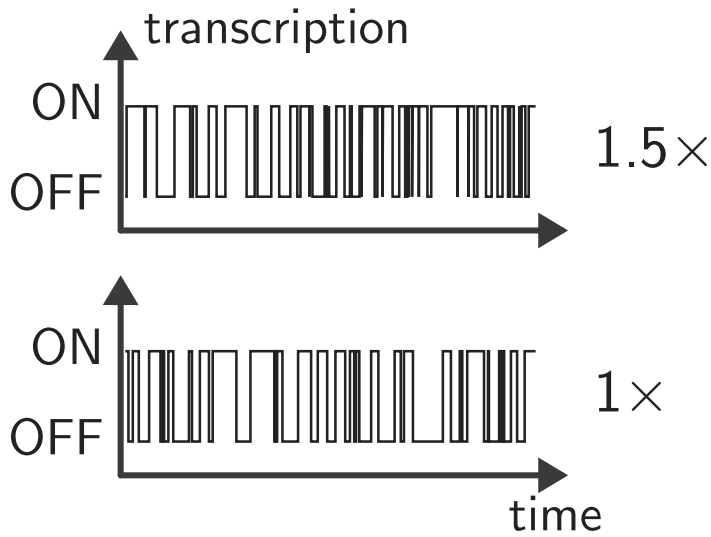

Conc.
difference
2×
1×
1/2×

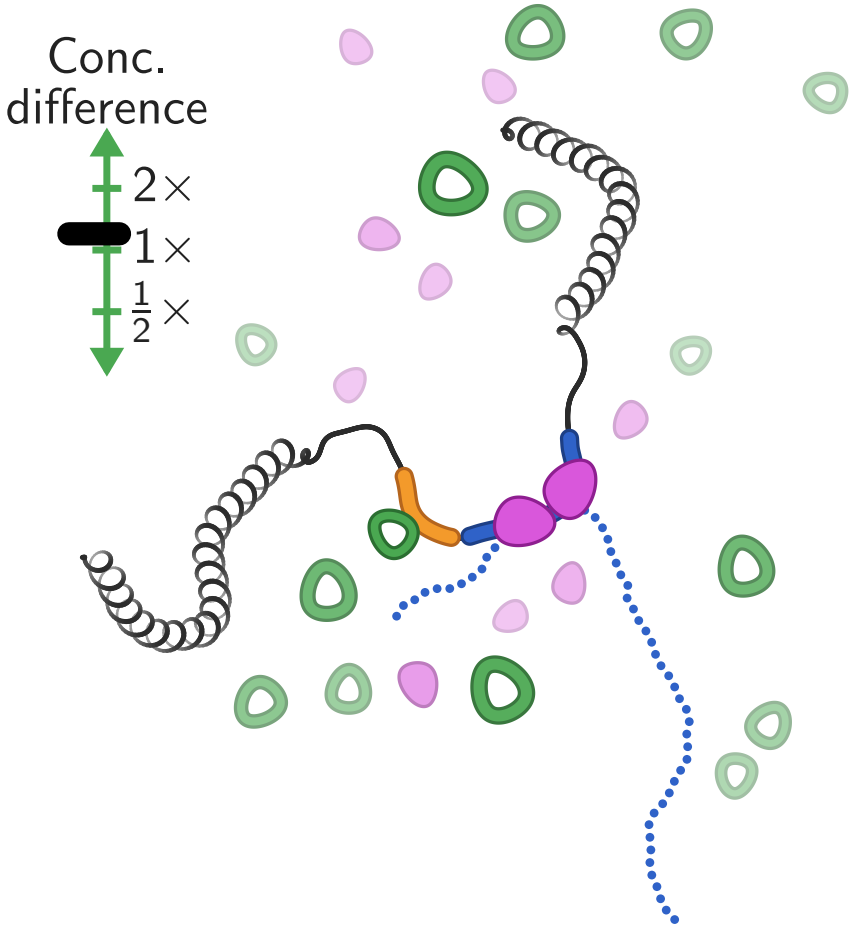

Transcription
OFF
$k_u$
$c\,k_b$
Transcription
ON

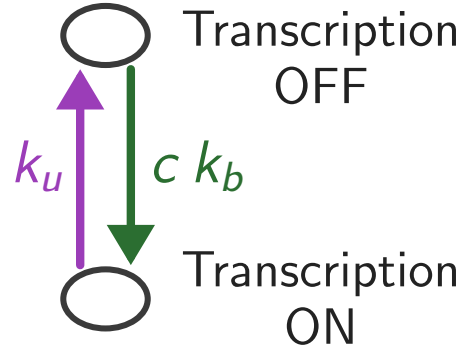

transcription
ON
OFF
time

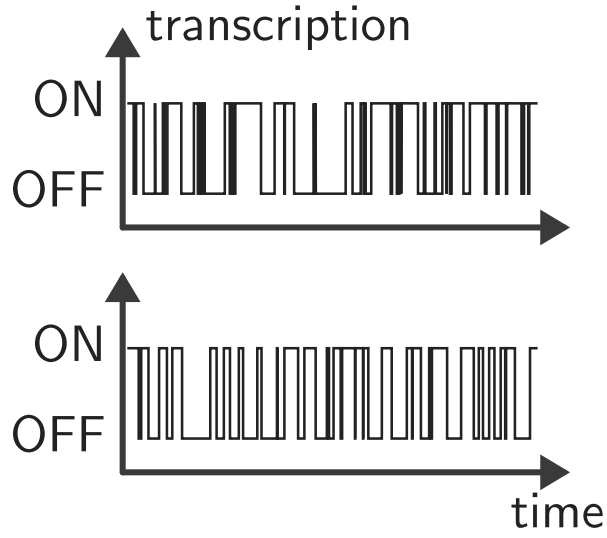

Conc.
difference
2×
1×
1/2×


Transcription
OFF
$k_u$
$c\,k_b$
Transcription
ON


transcription
ON
OFF
1.5×
ON
OFF
1×
time


Conc.
difference
2×
1×
1/2×


Transcription
OFF
$k_u$
$c\ k_b$
Transcription
ON


transcription
ON
OFF
ON
OFF
time

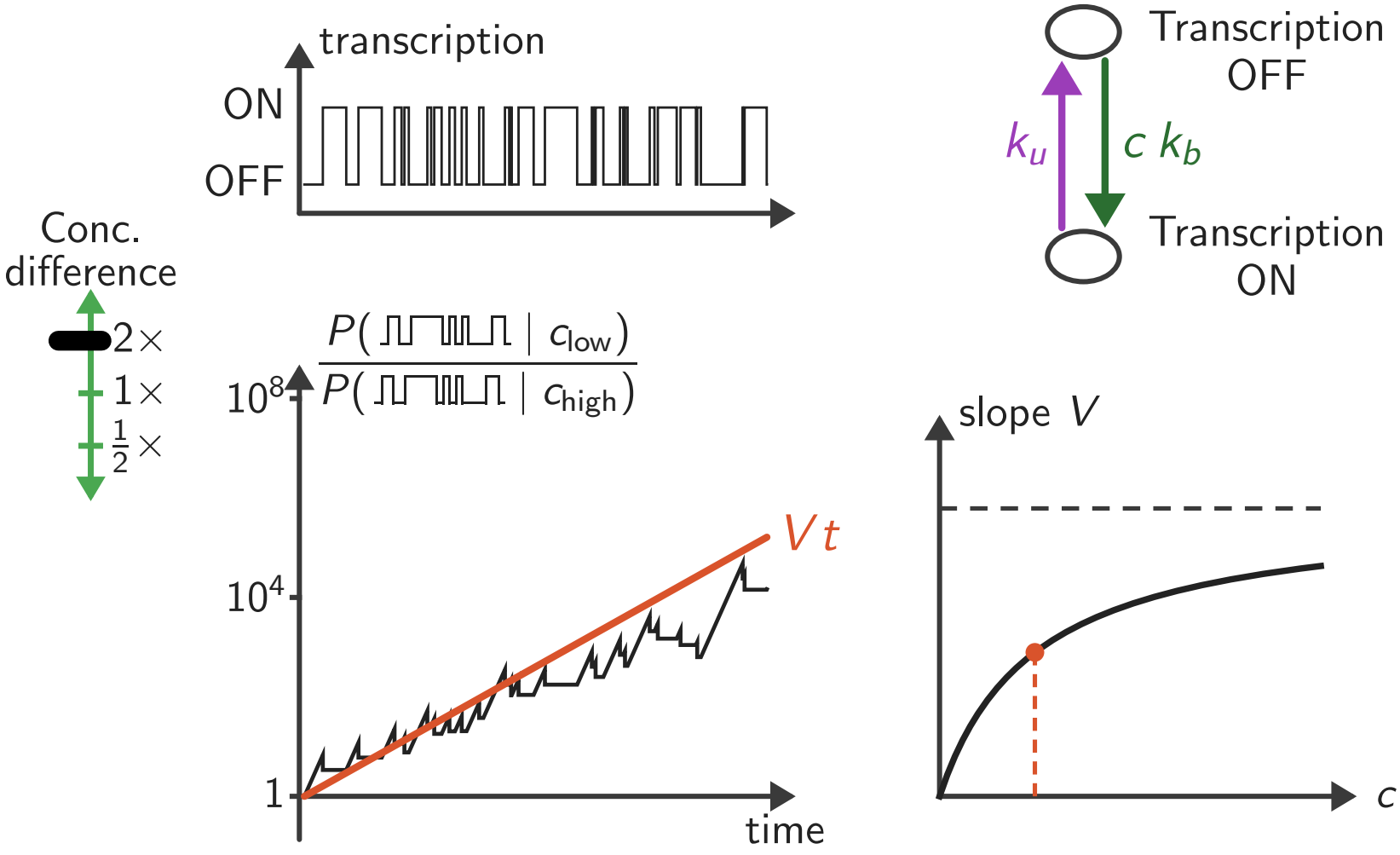
$$P(\text{  } \mid c_{\text{low}}) \overset{?}{>} P(\text{} \mid c_{\text{high}})$$

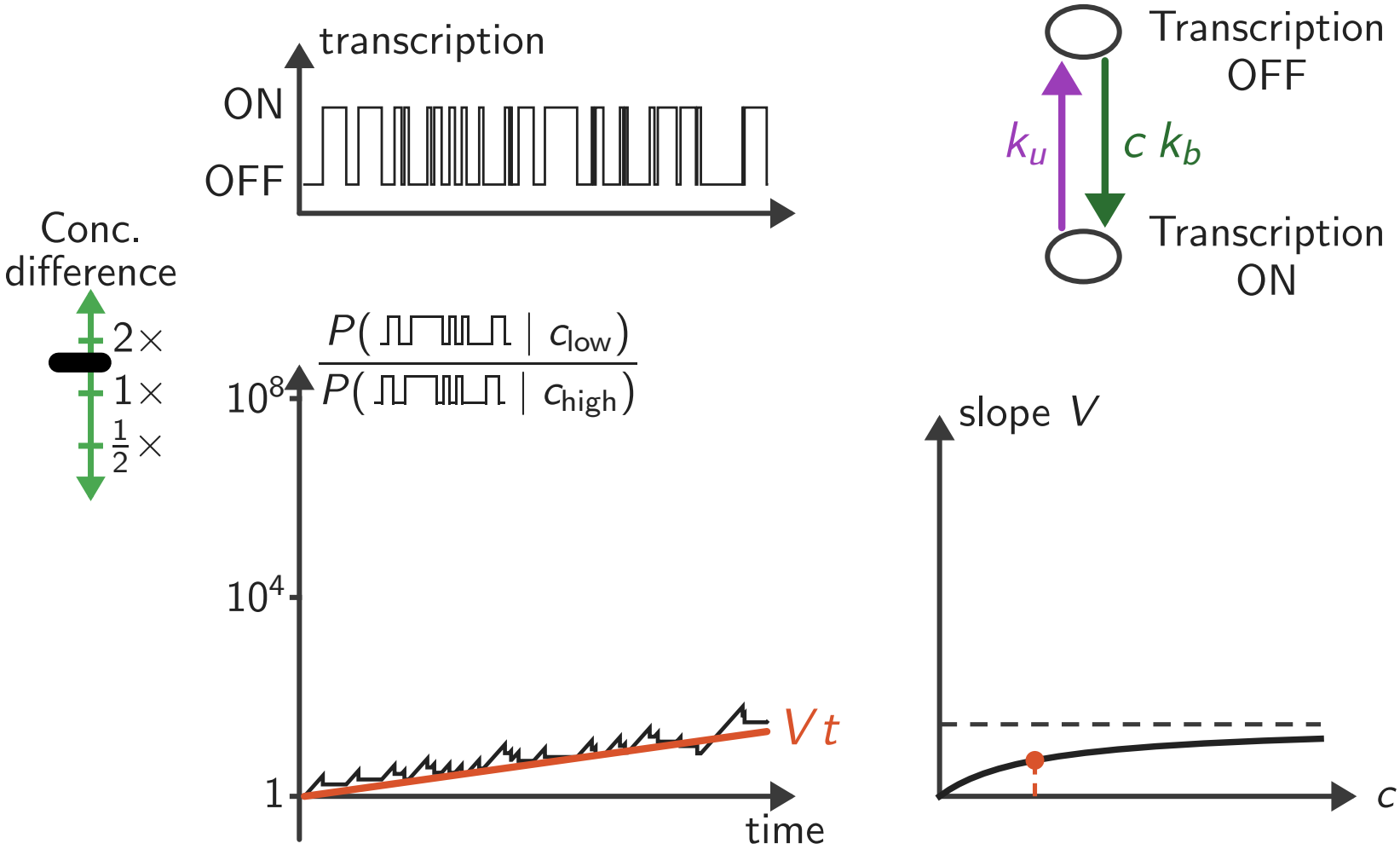
$$\frac{P(\text{} \mid c_{\text{low}})}{P(\text{} \mid c_{\text{high}})} > 1$$

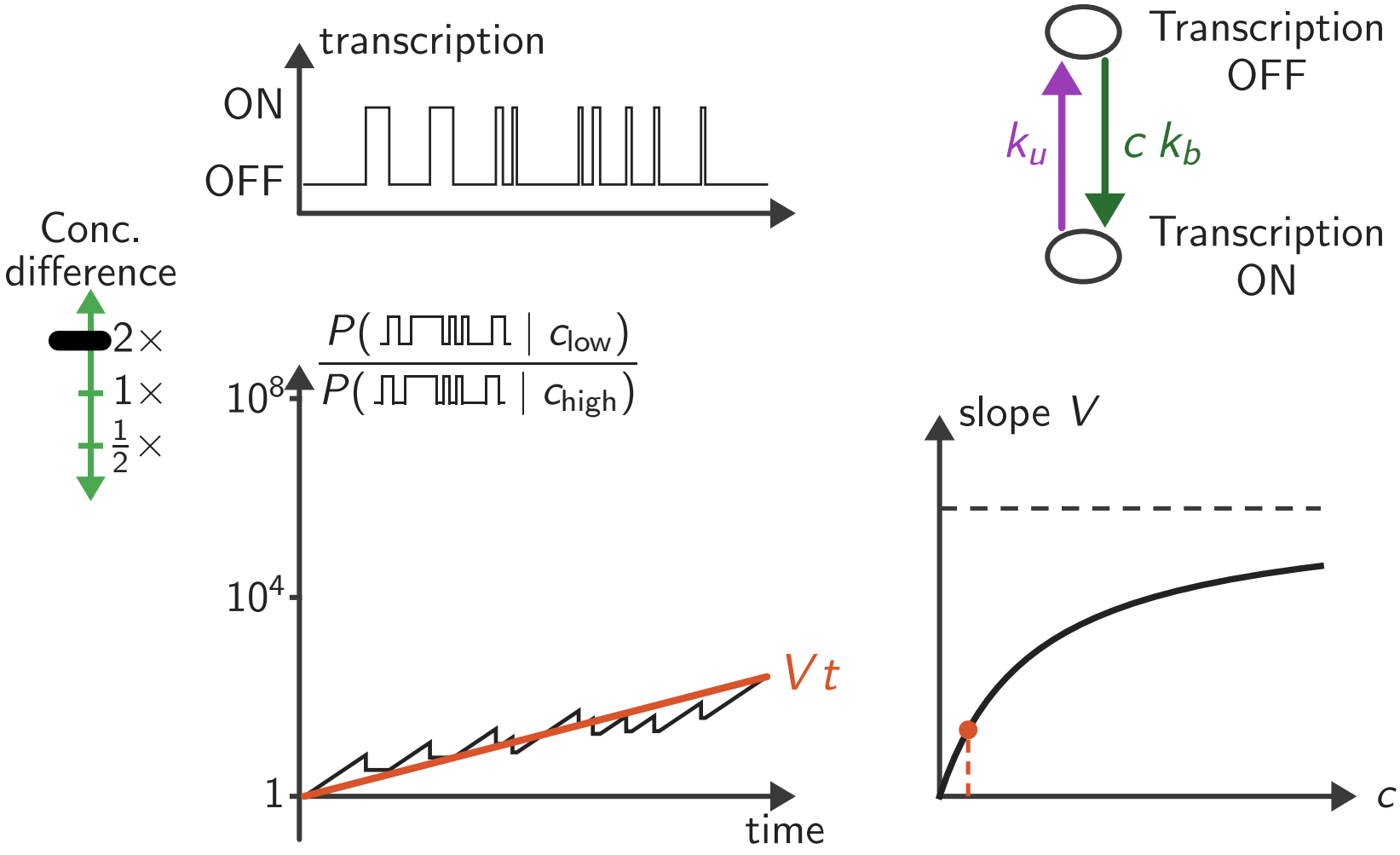
$$P(\ \text{}\ \mid c_{\text{low}}) \overset{?}{>} P(\ \text{}\ \mid c_{\text{high}})$$

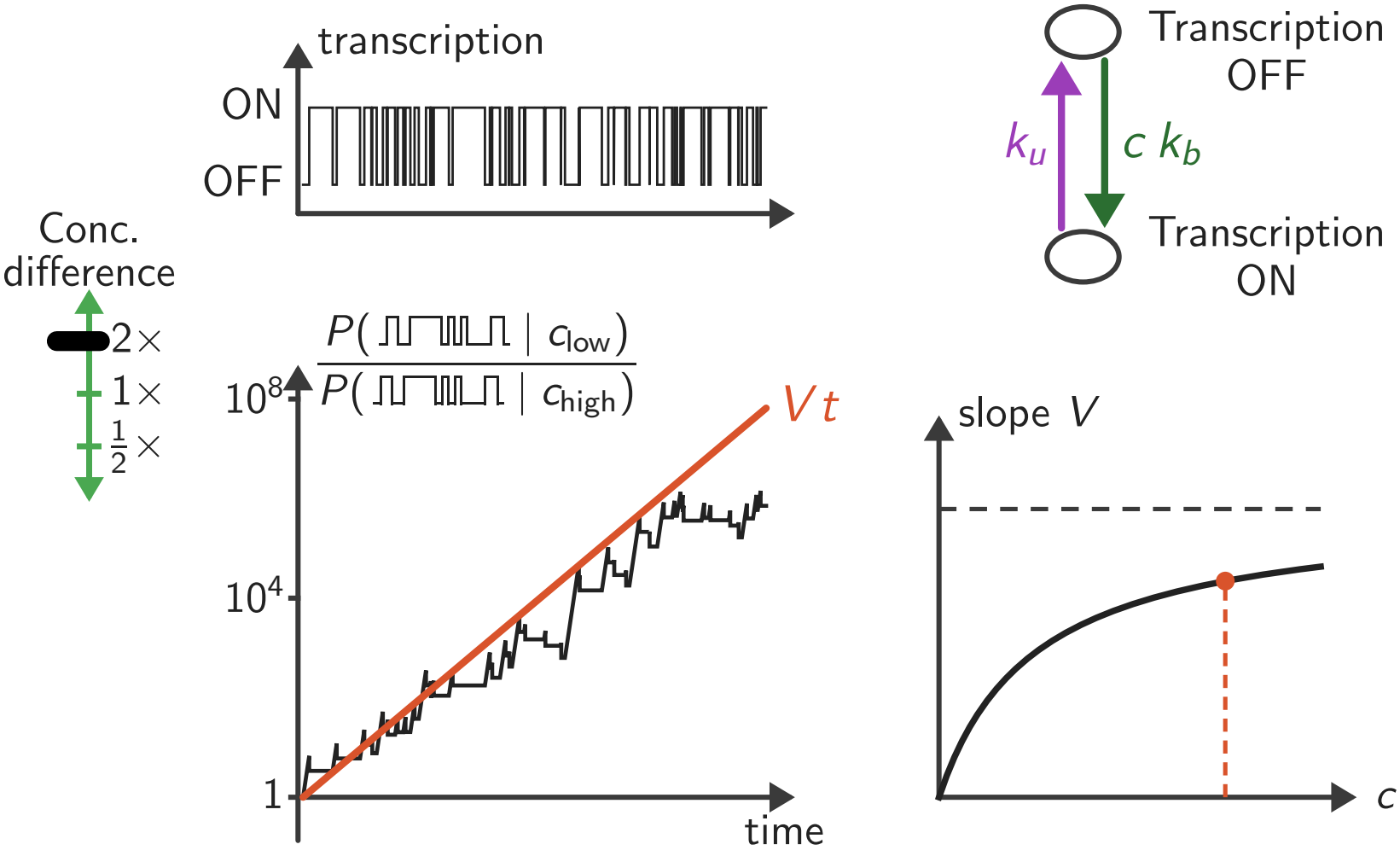
$$R(t) = \frac{P(\ \text{}\ \mid c_{\text{low}})}{P(\ \text{}\ \mid c_{\text{high}})} > 1$$

$$\log R(t) = V t + \sqrt{2D}\,\xi$$


transcription
ON
OFF
Conc. difference
2×
1×
1/2×
$\frac{P(\text{⊓⊓⊓} \mid c_{\text{low}})}{P(\text{⊓⊓⊓} \mid c_{\text{high}})}$
$10^8$
$10^4$
1
$Vt$
time
Transcription OFF
$k_u$
$c\,k_b$
Transcription ON
slope $V$
$c$


transcription
ON
OFF
Conc. difference
2×
1×
1/2×
$\frac{P(\text{⊓⊓⊓⊓} \mid c_{\text{low}})}{P(\text{⊓⊓⊓⊓} \mid c_{\text{high}})}$
$10^8$
$10^4$
1
$Vt$
time
Transcription OFF
$k_u$
$c\,k_b$
Transcription ON
slope $V$
$c$


transcription
ON
OFF
Conc. difference
2×
1×
1/2×
$\frac{P(\text{⊓⊓⊓} \mid c_{\text{low}})}{P(\text{⊓⊓⊓} \mid c_{\text{high}})}$
$10^8$
$10^4$
1
$Vt$
time
Transcription OFF
$k_u$
$c\,k_b$
Transcription ON
slope $V$
$c$


transcription
ON
OFF
Conc. difference
2×
1×
1/2×
$\frac{P(\text{⊓⊓⊓} \mid c_{\text{low}})}{P(\text{⊓⊓⊓} \mid c_{\text{high}})}$
$10^8$
$10^4$
1
time
$Vt$
Transcription OFF
$k_u$
$c\,k_b$
Transcription ON
slope $V$
$c$

$$V = \frac{1}{2}\left(\frac{\delta c}{c}\right)^2 \underbrace{\dot{\mathcal{F}}(c)}_{\text{Fisher information rate}}$$

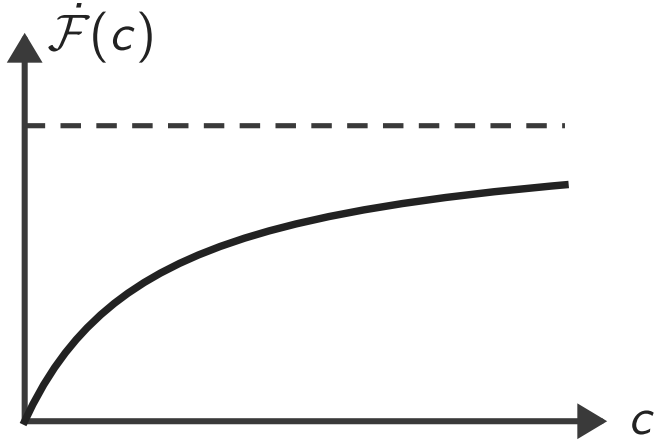
$$V = \frac{1}{2}\left(\frac{\delta c}{c}\right)^2 \underbrace{\dot{\mathcal{F}}(c)}_{\text{Fisher information rate}}$$

$$\dot{\mathcal{F}}(c) = -\lim_{t\to\infty} \frac{1}{t}\left\langle \frac{\partial^2 \ln P(\mathcal{O}(t) \mid c)}{(\partial \ln c)^2} \right\rangle_c$$

$$V = \frac{1}{2}\left(\frac{\delta c}{c}\right)^2 \underbrace{\dot{\mathcal{F}}(c)}_{\text{Fisher information rate}}$$

$$\dot{\mathcal{F}}(c) = -\lim_{t\to\infty} \frac{1}{t}\left\langle \frac{\partial^2 \ln P(\mathcal{O}(t)\,|\,c)}{(\partial \ln c)^2} \right\rangle_c$$

$$= \frac{1}{\tau_{\text{burst}}}$$




$c\,k_b$: diffusion-limited arrival of molecules

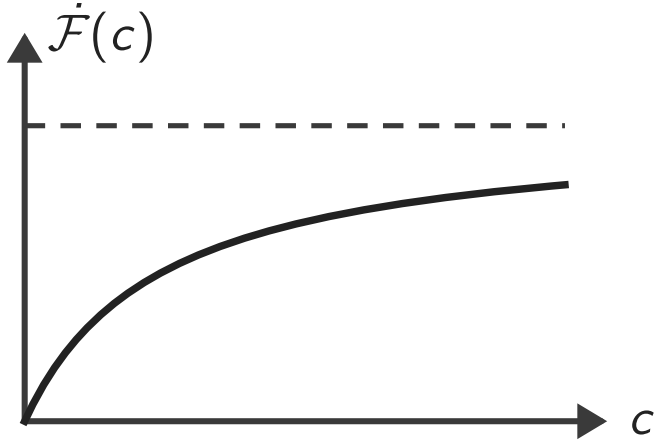
$k_b = 4aD$

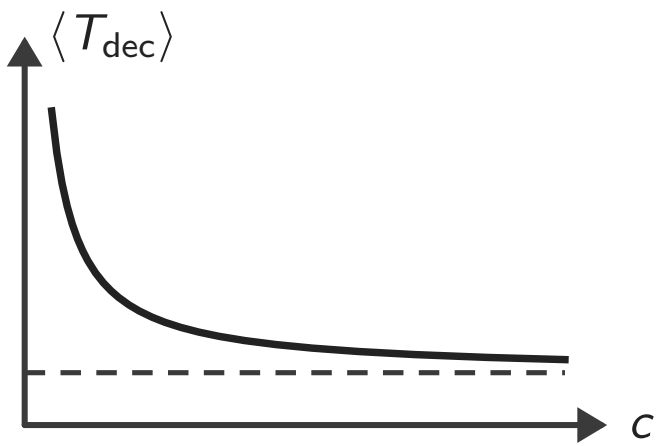
$$V = \frac{1}{2}\left(\frac{\delta c}{c}\right)^2 \underbrace{\dot{\mathcal{F}}(c)}_{\text{Fisher information rate}}$$

$$\dot{\mathcal{F}}(c) = -\lim_{t\to\infty} \frac{1}{t}\left\langle \frac{\partial^2 \ln P(\mathcal{O}(t)\,|\,c)}{(\partial \ln c)^2}\right\rangle_c$$

$$= \frac{1}{\tau_{\text{burst}}}$$

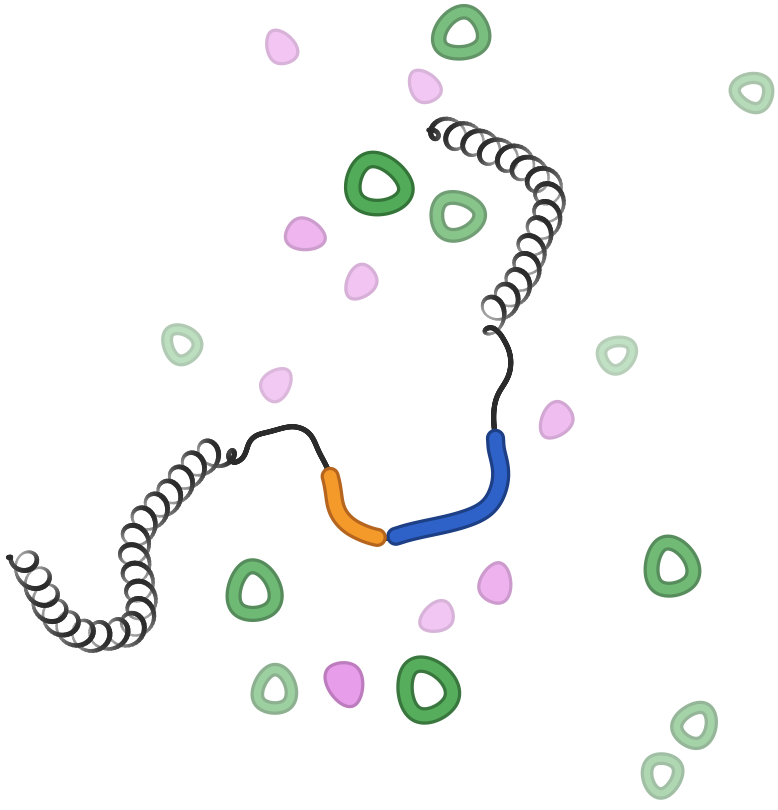
$$\langle T_{\text{dec}}\rangle = \frac{K}{V}\tanh\left(\frac{K}{2}\right)$$

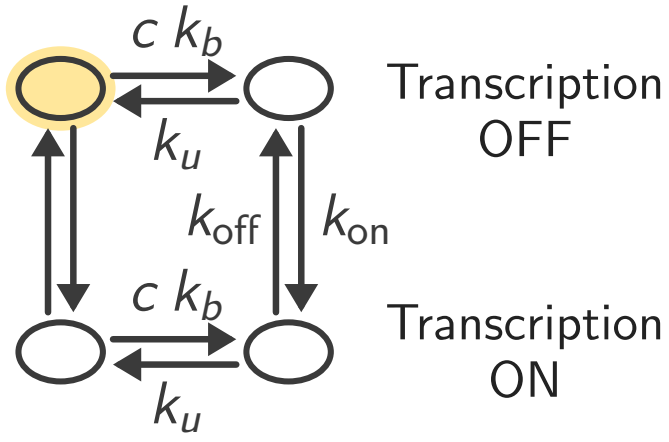


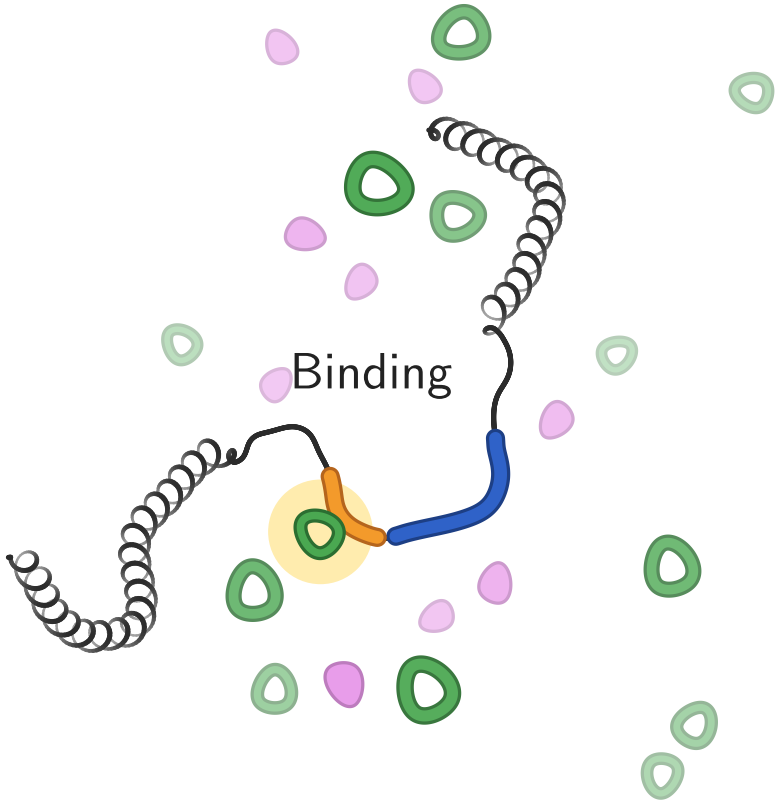



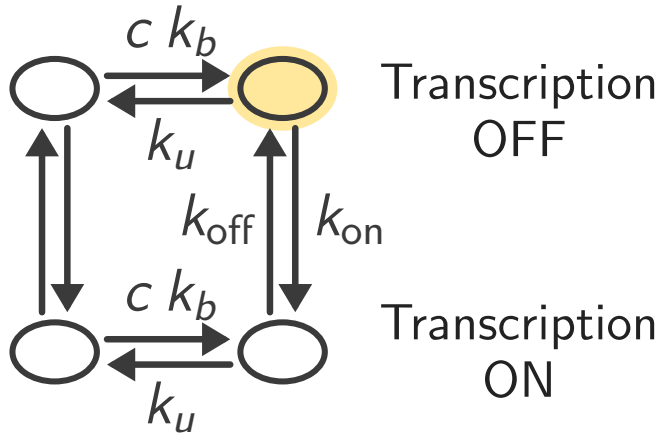

$c\,k_b$: diffusion-limited arrival of molecules

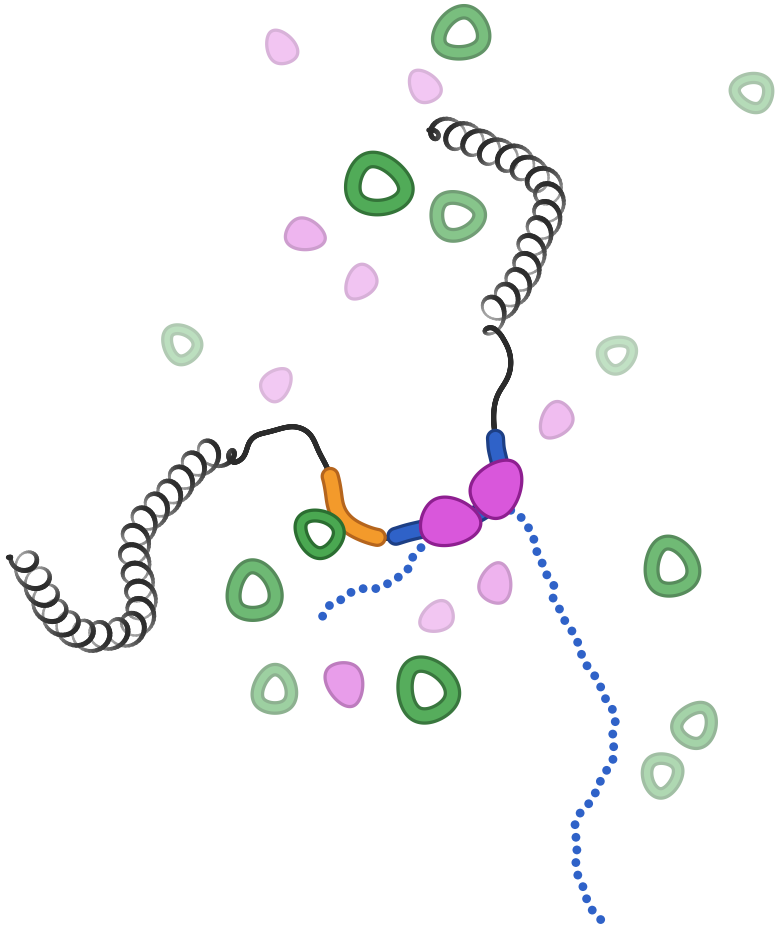
$k_b = 4aD$

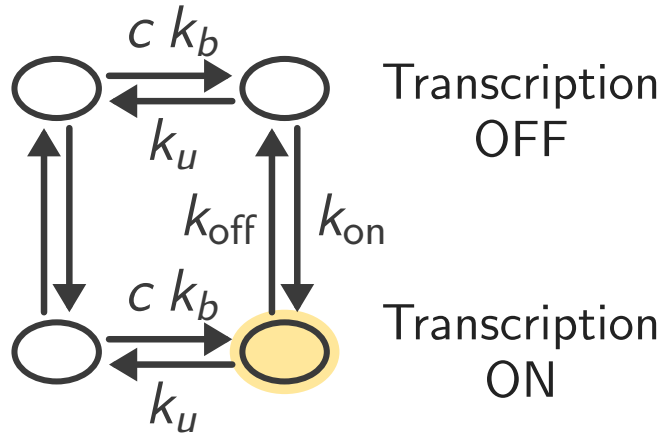


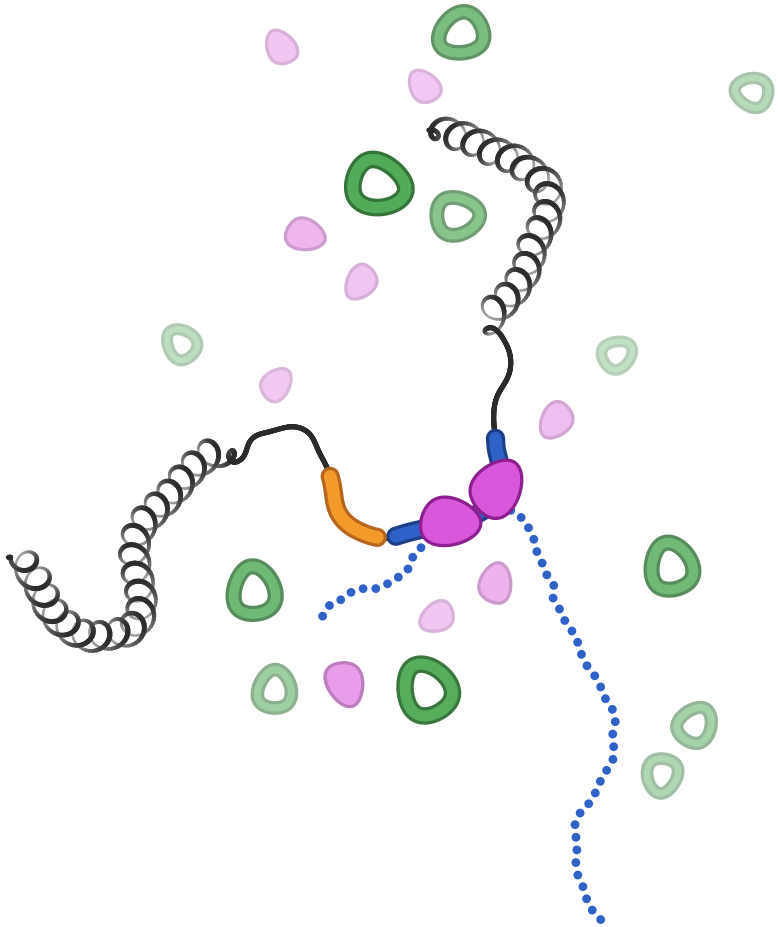

$c\ k_b$
$k_u$
$k_{\text{off}}$
$k_{\text{on}}$
$c\ k_b$
$k_u$
Transcription OFF
Transcription ON

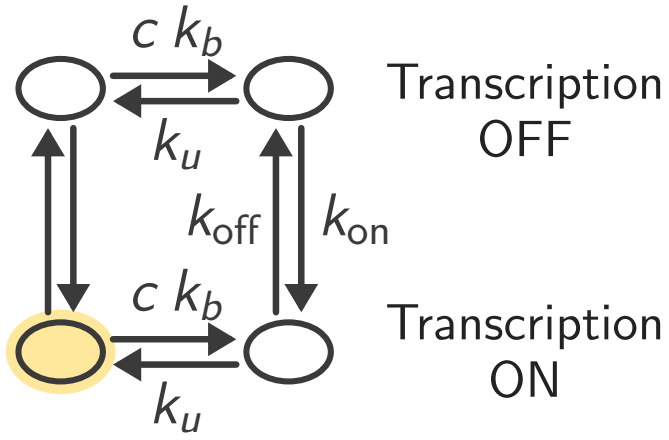

Binding

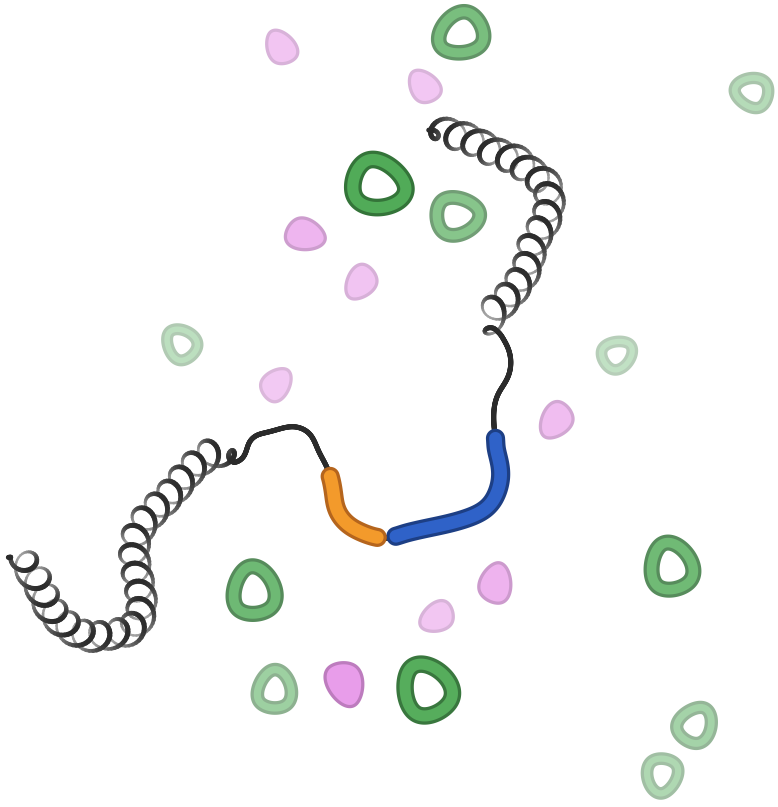

c $k_b$
$k_u$
Transcription OFF
$k_{off}$
$k_{on}$
c $k_b$
$k_u$
Transcription ON

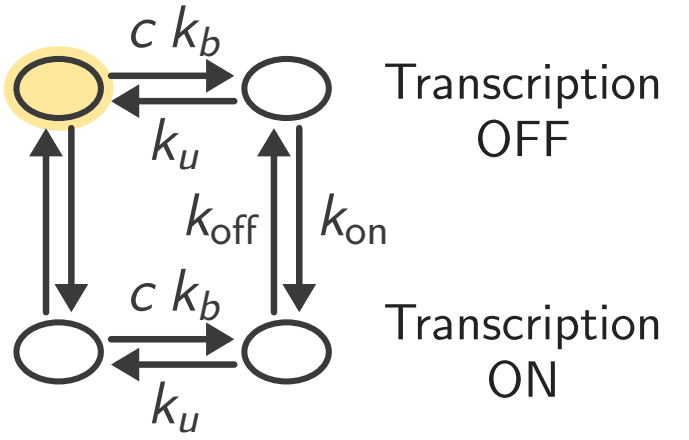


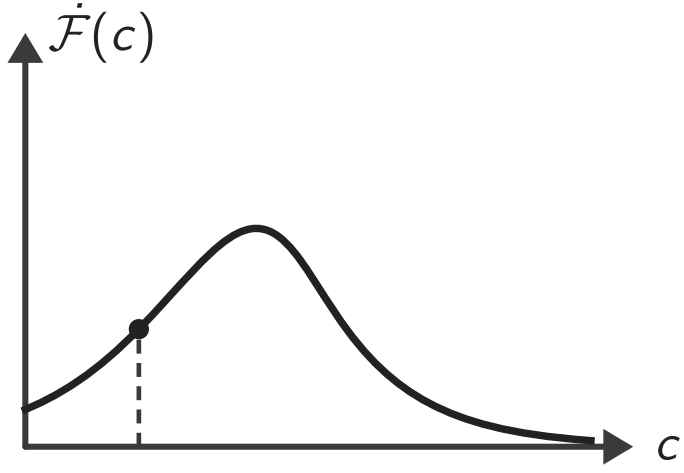

$c\ k_b$
$k_u$
Transcription
OFF
$k_{\text{off}}$
$k_{\text{on}}$
$c\ k_b$
$k_u$
Transcription
ON

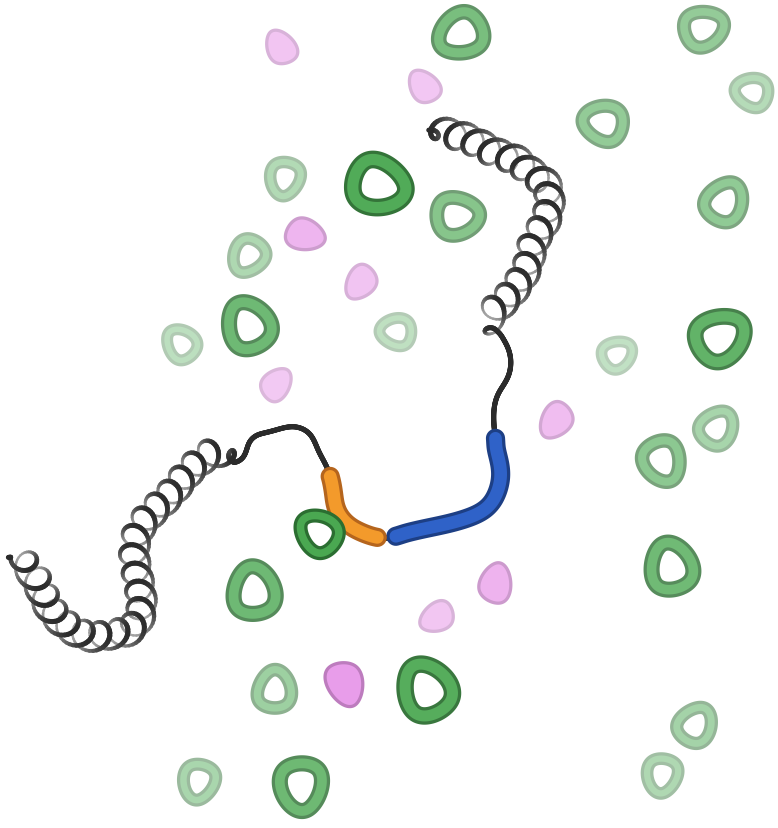


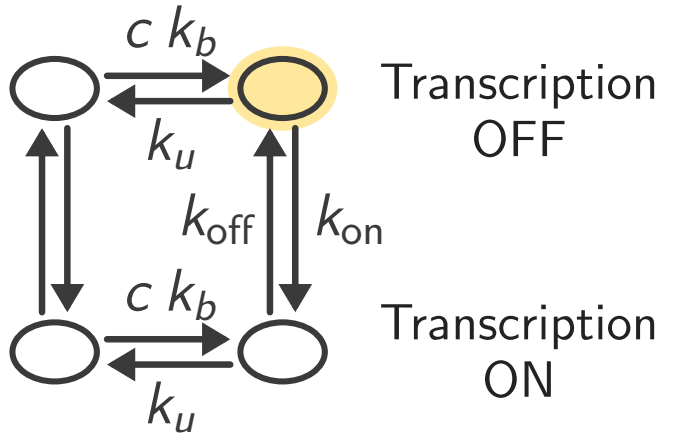

$c\ k_b$
$k_u$
Transcription
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$k_{\text{off}}$
$k_{\text{on}}$
$c\ k_b$
$k_u$
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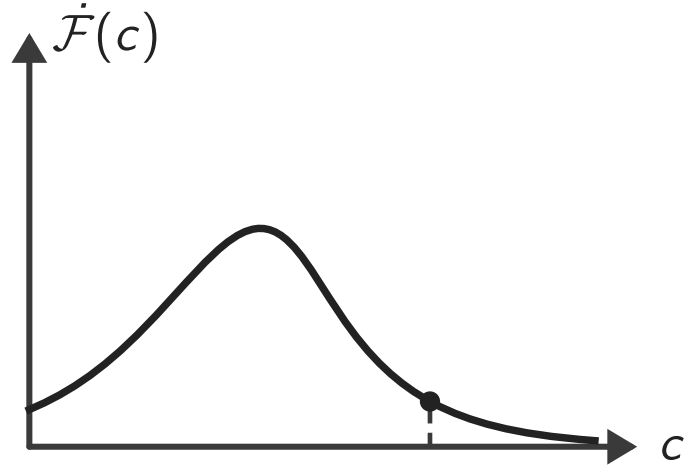


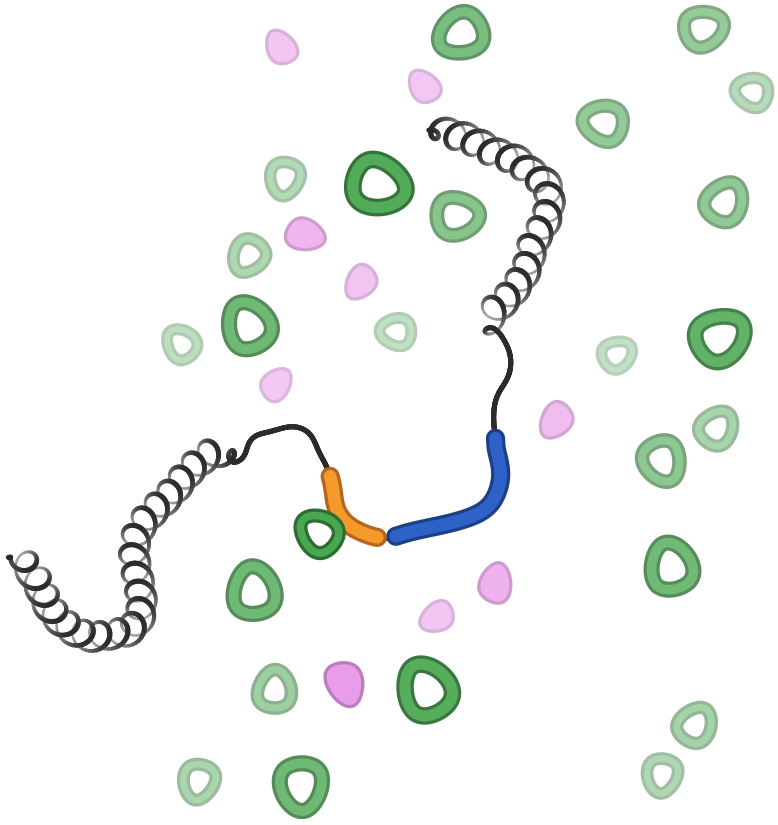

$c\ k_b$
$k_u$
Transcription
OFF
$k_{\text{off}}$
$k_{\text{on}}$
$c\ k_b$
$k_u$
Transcription
ON

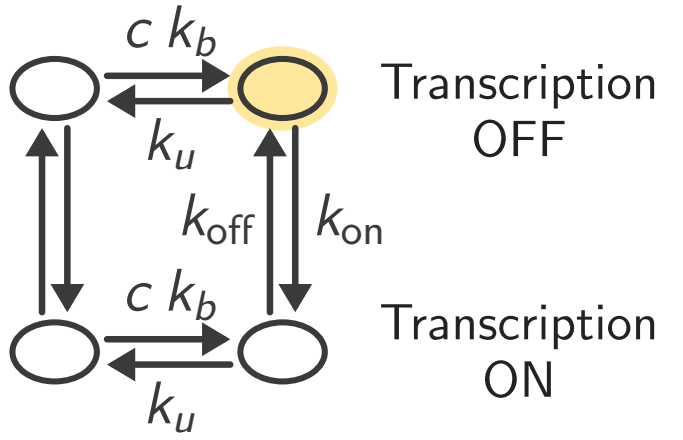

$\dot{\mathcal{F}}(c)$
$c$

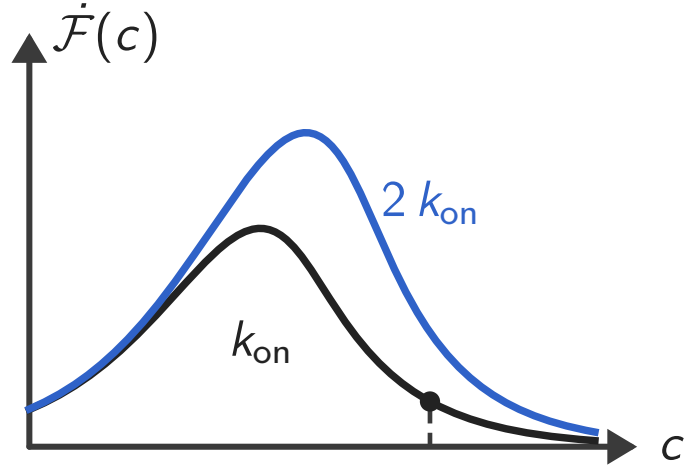


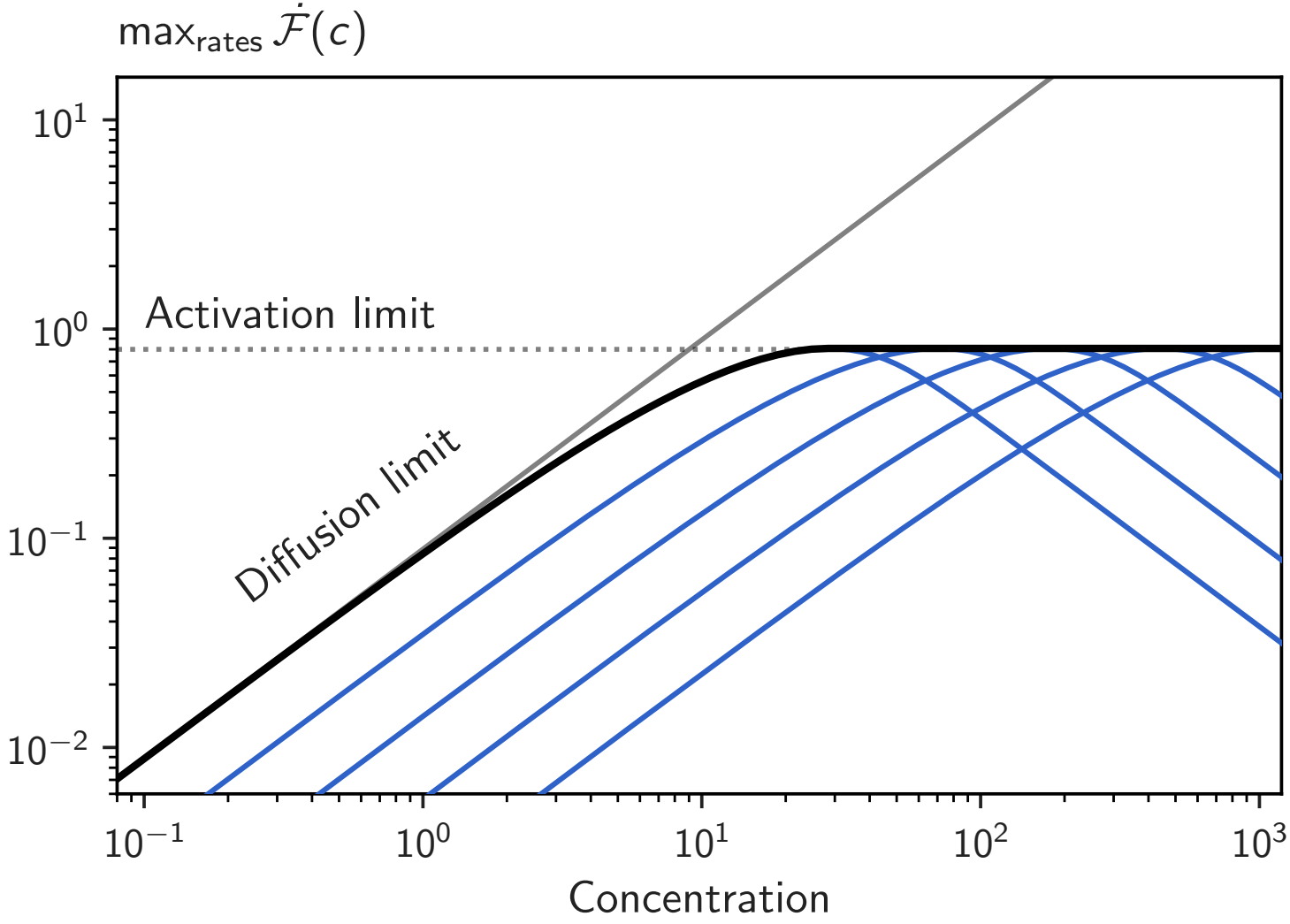

$c\ k_b$
$k_u$
Transcription
OFF
$k_{\text{off}}$
$k_{\text{on}}$
$c\ k_b$
$k_u$
Transcription
ON

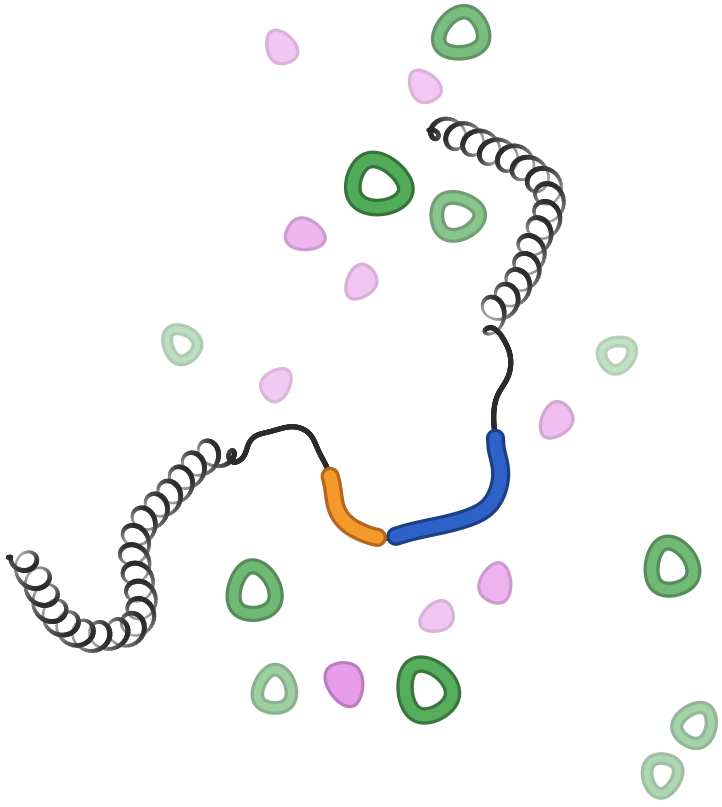

$\dot{\mathcal{F}}(c)$
$c$

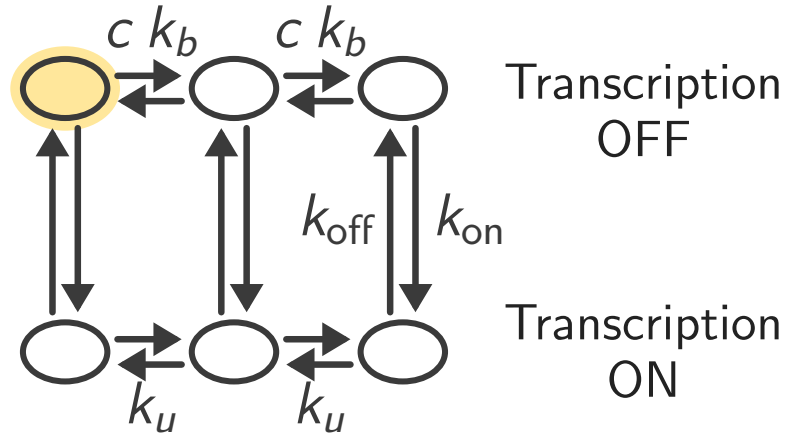


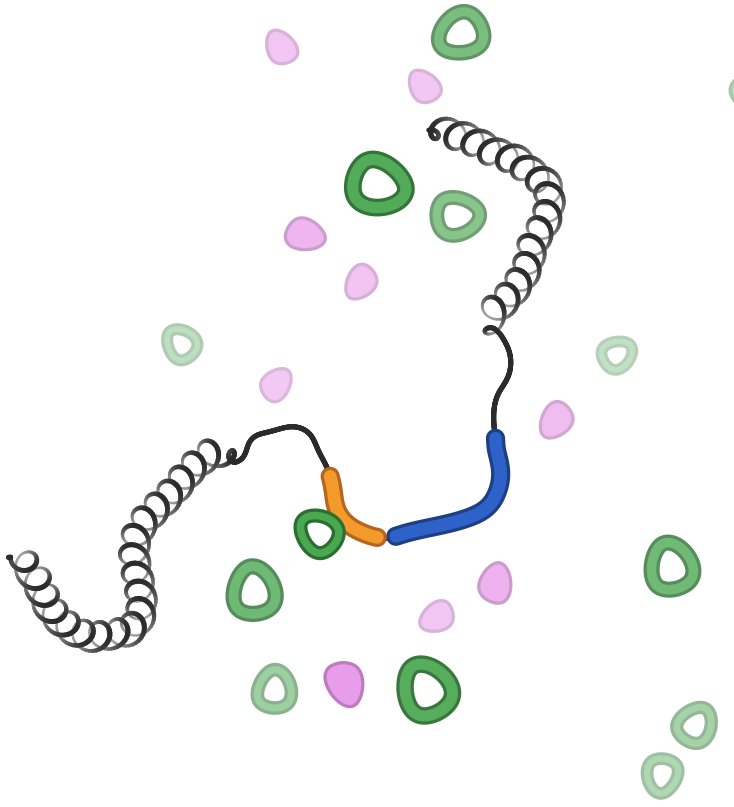

$c\ k_b$
$k_u$
Transcription
OFF
$k_{\text{off}}$
$k_{\text{on}}$
$c\ k_b$
$k_u$
Transcription
ON

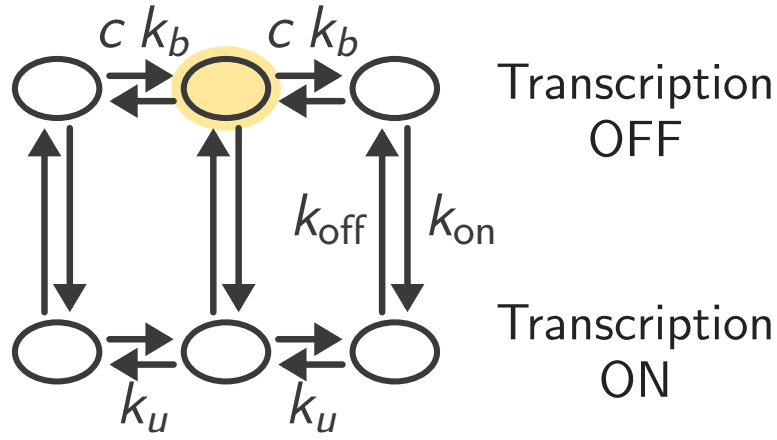

$\dot{\mathcal{F}}(c)$
$2\ k_{\text{on}}$
$k_{\text{on}}$
$c$

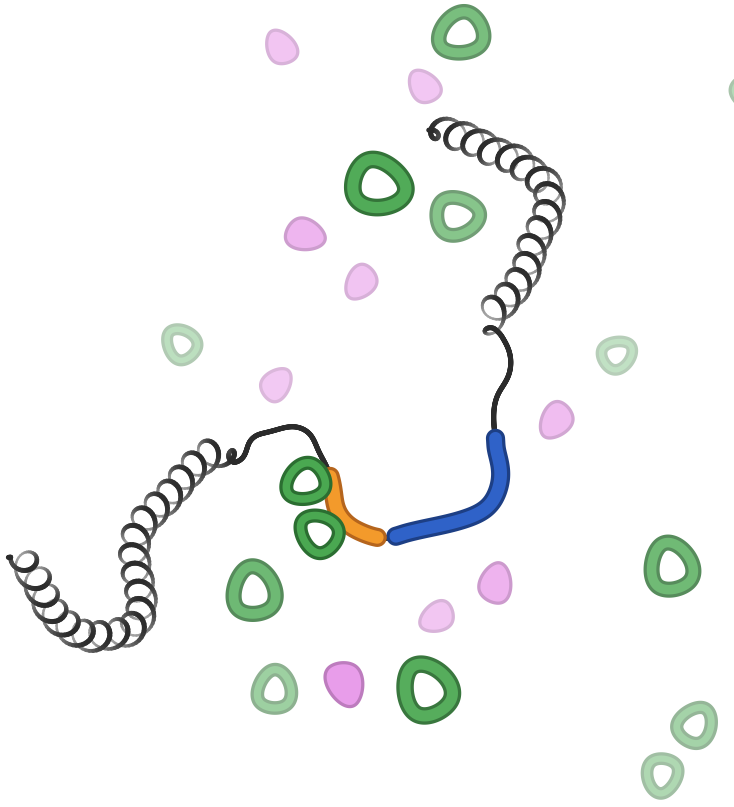

$\max_{\text{rates}} \dot{\mathcal{F}}(c)$
$10^1$
$10^0$
$10^{-1}$
$10^{-2}$
Activation limit
Diffusion limit
$10^{-1}$
$10^0$
$10^1$
$10^2$
$10^3$
Concentration

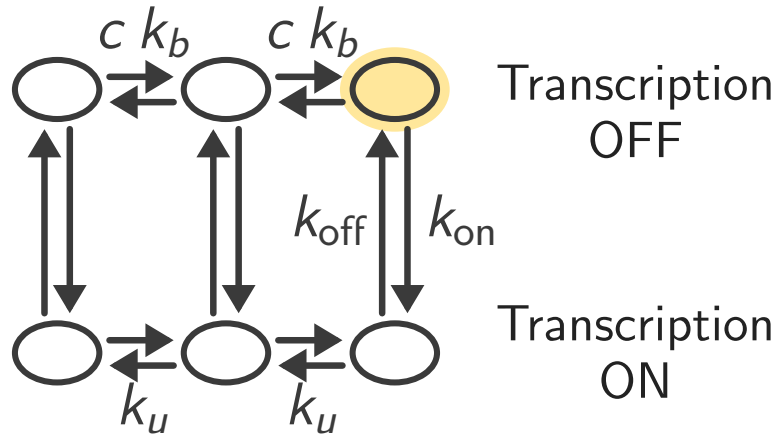


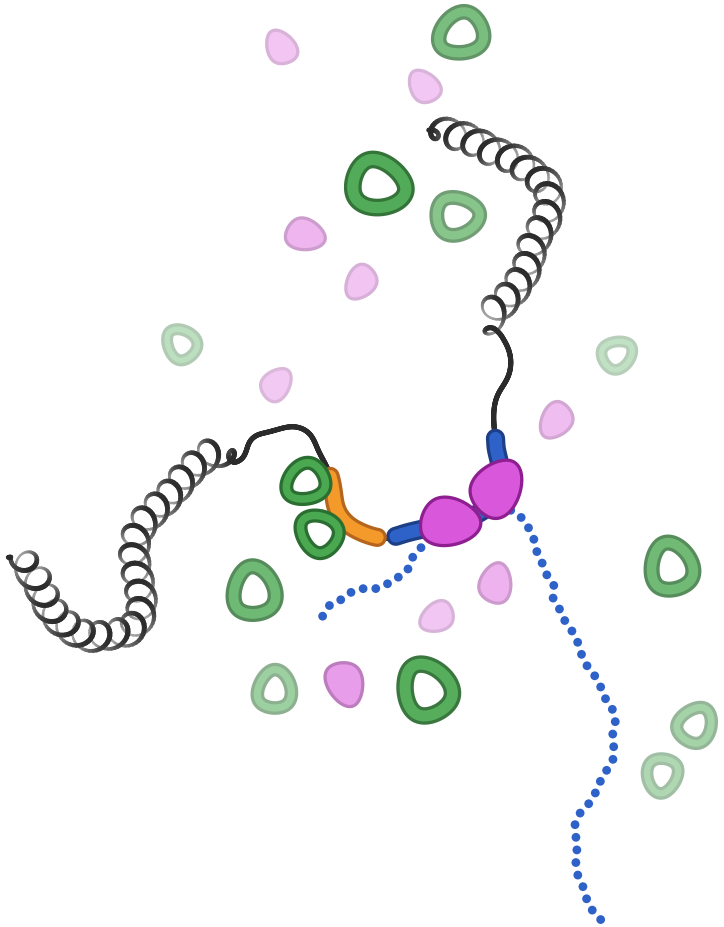

c $k_b$
c $k_b$
Transcription OFF
$k_{off}$
$k_{on}$
Transcription ON
$k_u$
$k_u$

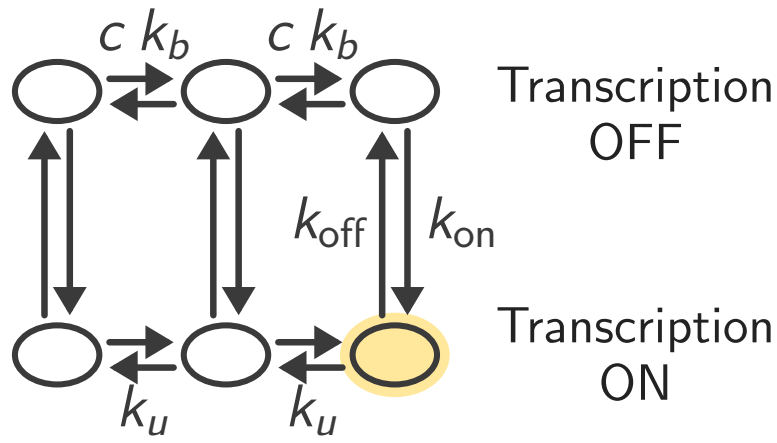


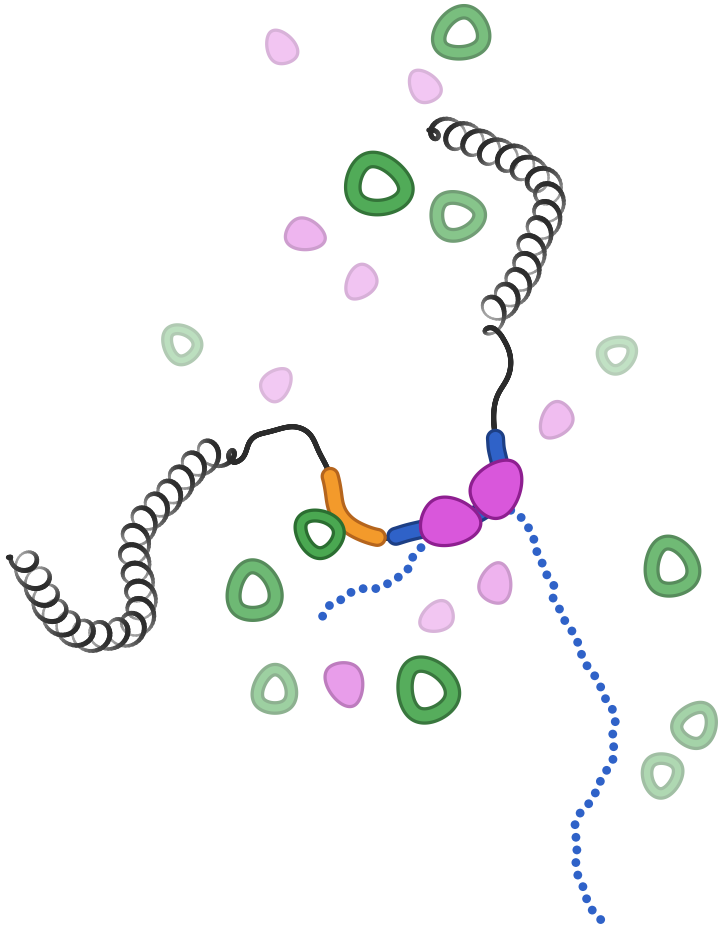

$c\ k_b$
$c\ k_b$
Transcription OFF
$k_{\text{off}}$
$k_{\text{on}}$
Transcription ON
$k_u$
$k_u$

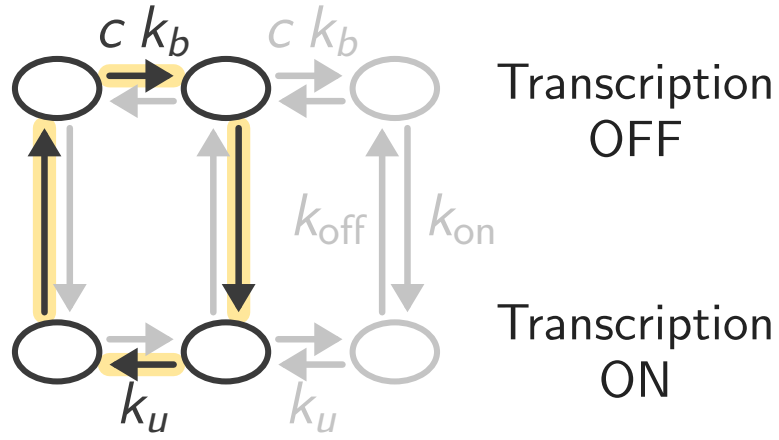


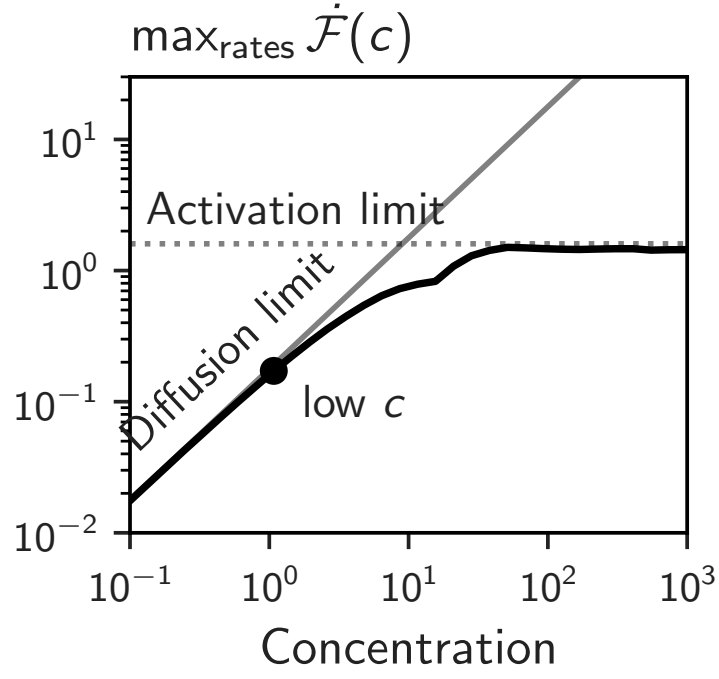

$c\ k_b$
$c\ k_b$
Transcription OFF
$k_{\text{off}}$
$k_{\text{on}}$
Transcription ON
$k_u$
$k_u$

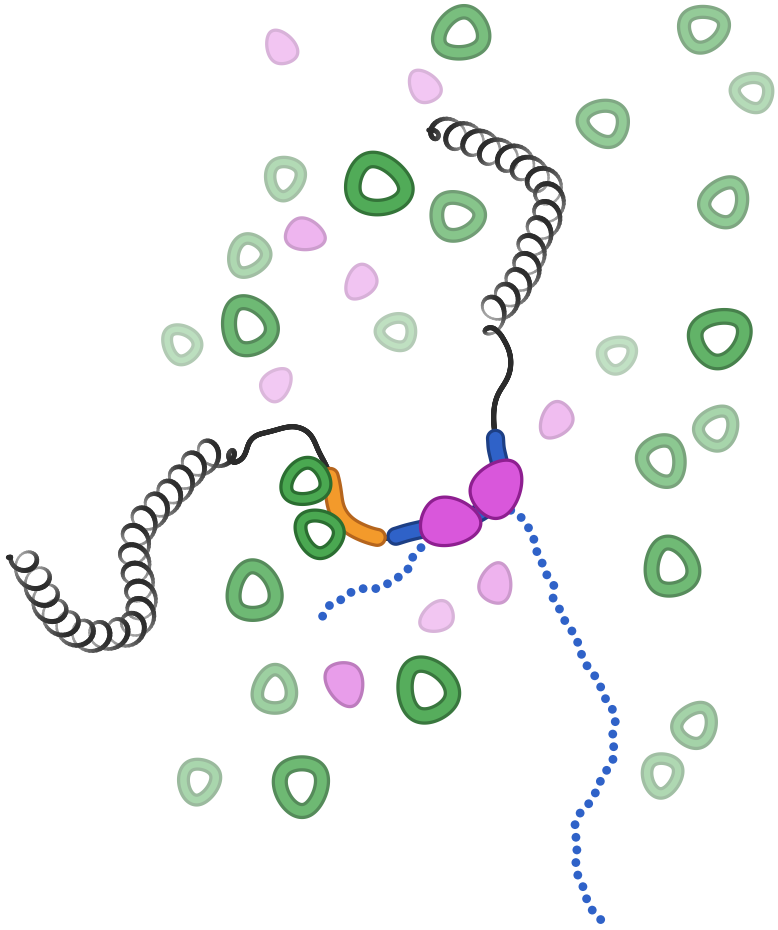


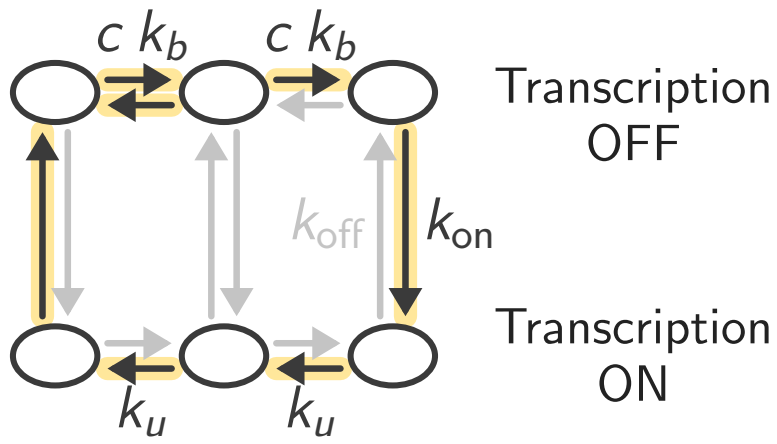

c $k_b$
c $k_b$
Transcription OFF
$k_{off}$
$k_{on}$
Transcription ON
$k_u$
$k_u$

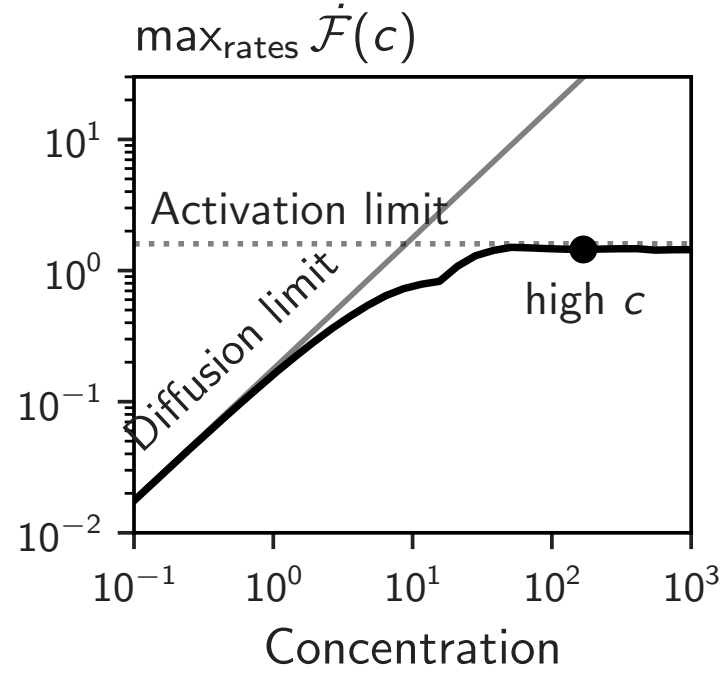


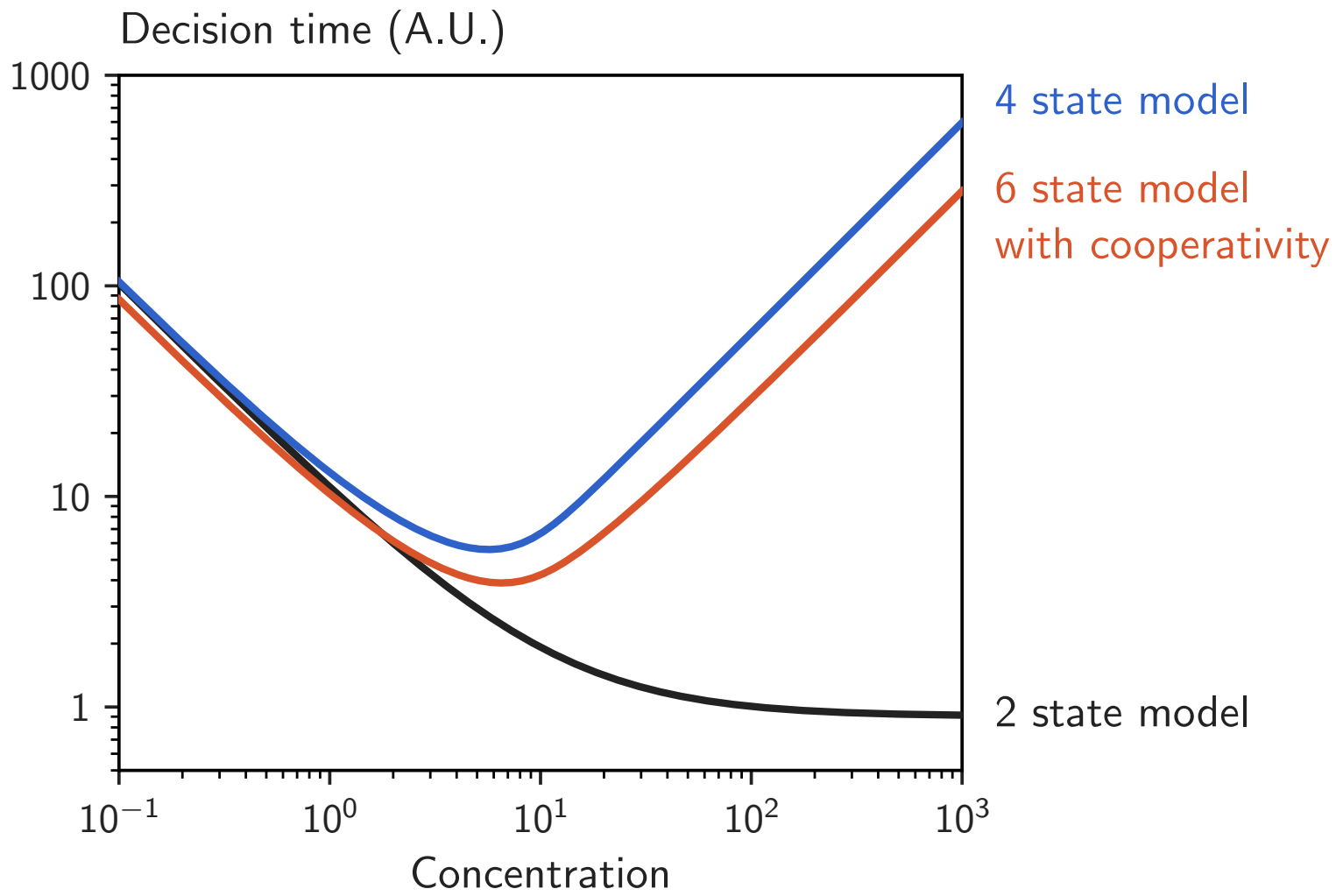

$c\ k_b$
$c\ k_b$
Transcription OFF
$k_{\text{off}}$
$k_{\text{on}}$
Transcription ON
$k_u$
$k_u$


$\max_{\text{rates}} \dot{\mathcal{F}}(c)$
$10^1$
$10^0$
$10^{-1}$
$10^{-2}$
Activation limit
Diffusion limit
low $c$
$10^{-1}$
$10^0$
$10^1$
$10^2$
$10^3$
Concentration

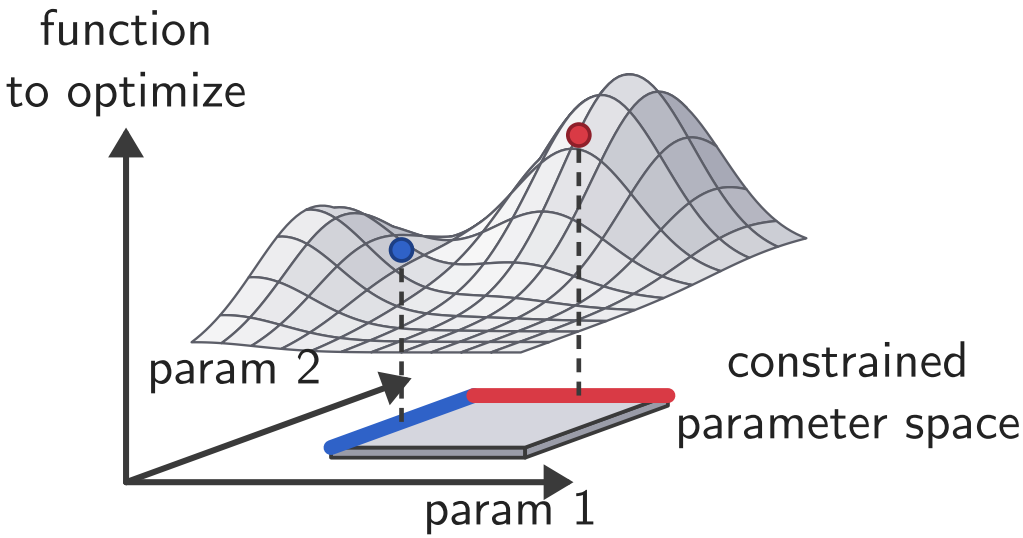



c $k_b$
c $k_b$
Transcription OFF
$k_{off}$
$k_{on}$
Transcription ON
$k_u$
$k_u$

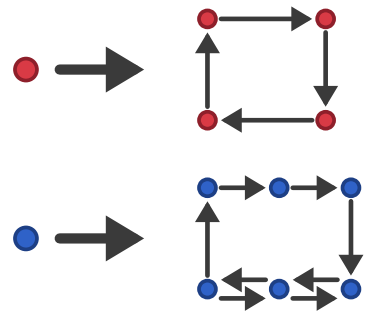

$\max_{\text{rates}} \dot{\mathcal{F}}(c)$
$10^1$
$10^0$
$10^{-1}$
$10^{-2}$
Activation limit
Diffusion limit
high $c$
$10^{-1}$
$10^0$
$10^1$
$10^2$
$10^3$
Concentration

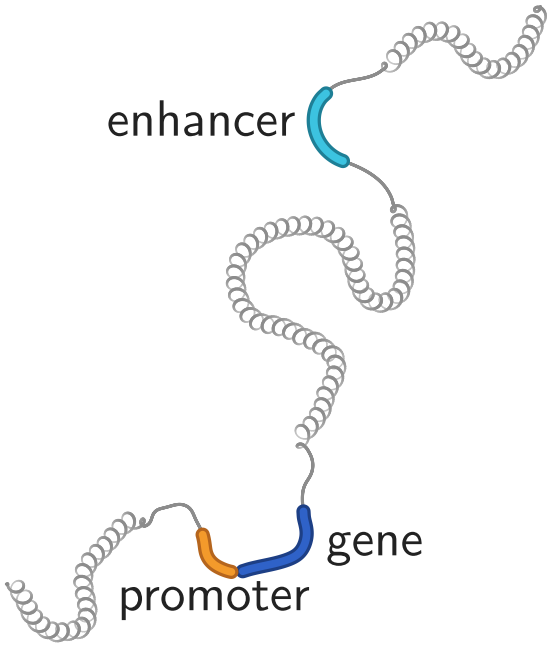

Decision time (A.U.)
1000
100
10
1
$10^{-1}$
$10^{0}$
$10^{1}$
$10^{2}$
$10^{3}$
Concentration
4 state model
6 state model
with cooperativity
2 state model

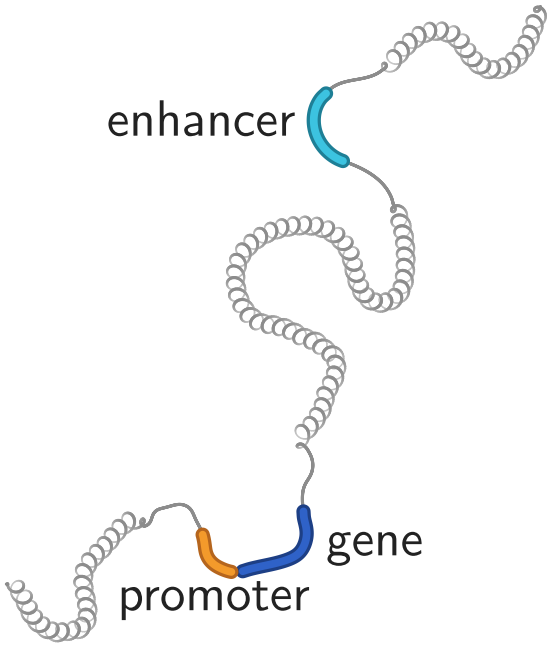
# Takeaway

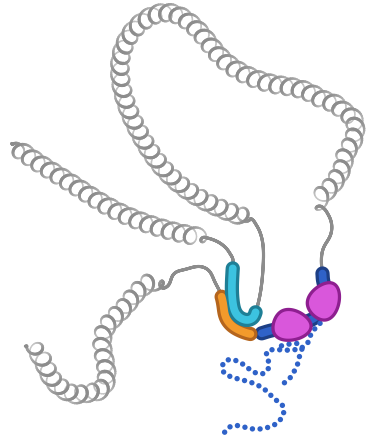


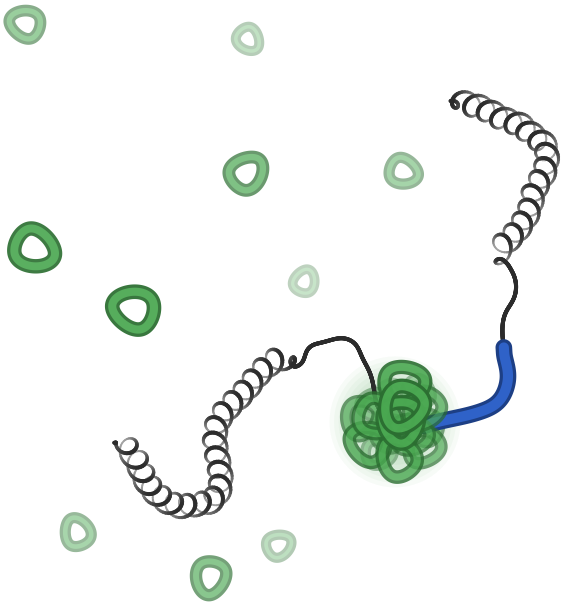

same optimized function
different mechanisms

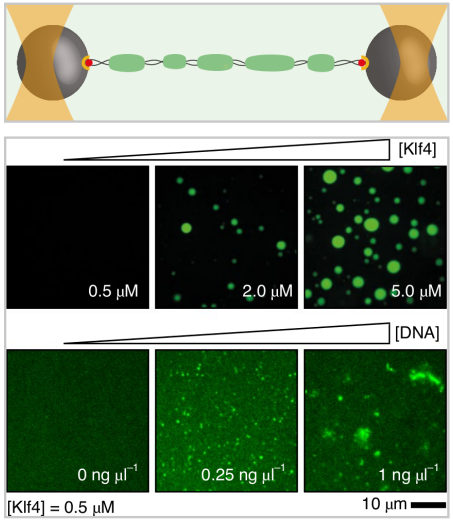



enhancer
gene
promoter

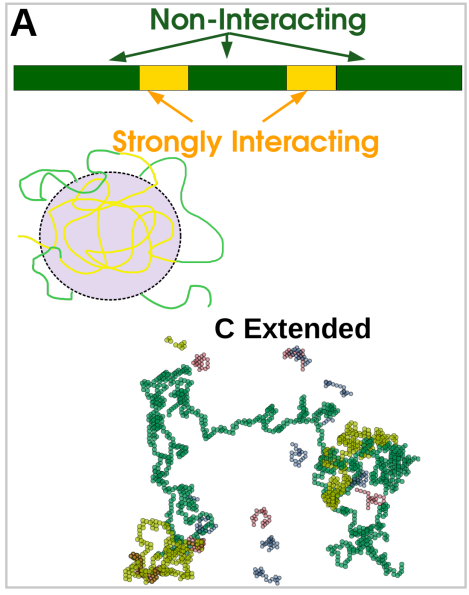

enhancer
gene
promoter



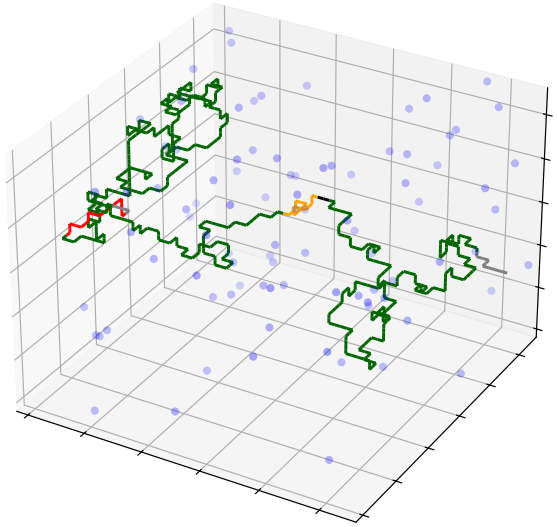


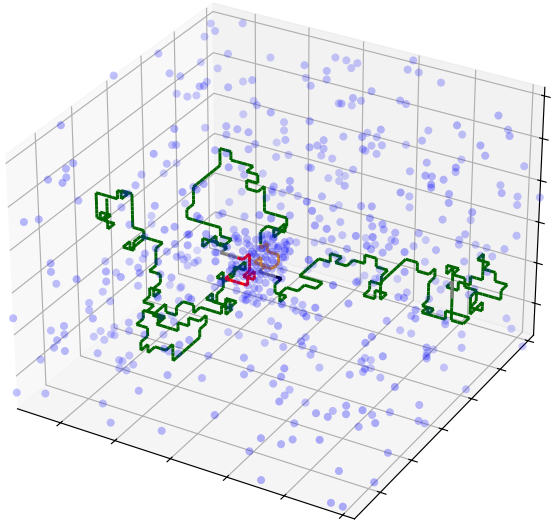


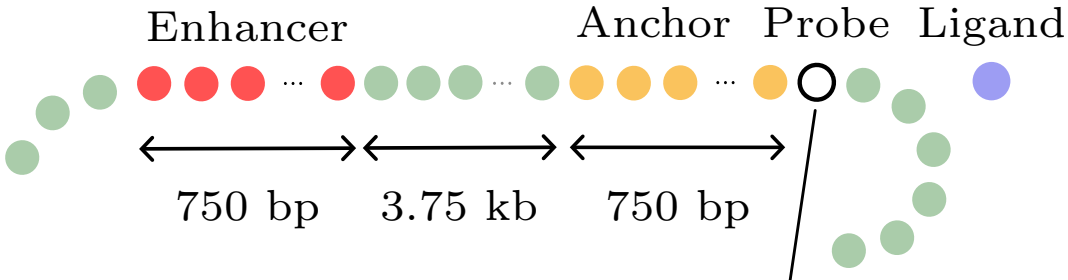

*Sequence-dependent surface condensation of a pioneer transcription factor on DNA*
Morin et al., 2022



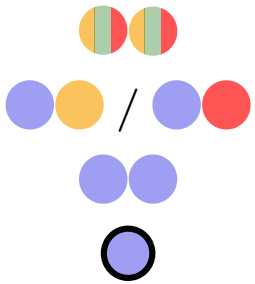

*Protein-DNA Co-condensation is Prewetting to a Collapsed Polymer*
Rouches & Machta, 2025








Particle interactions



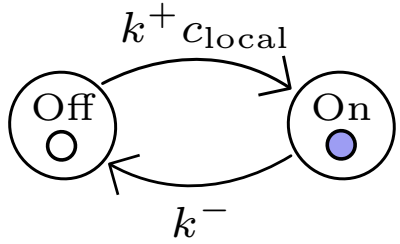
$v = +0.5$

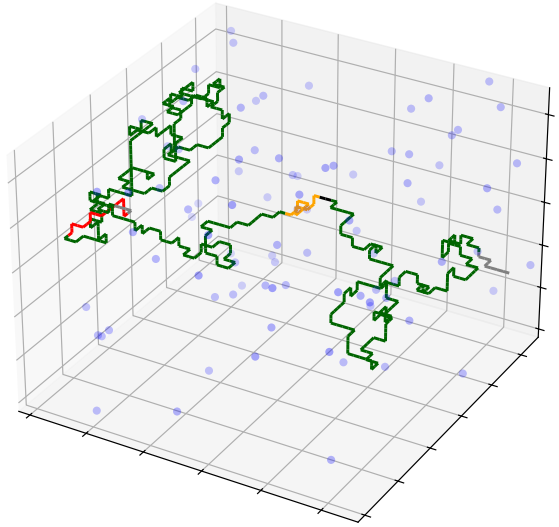
$\eta = -0.8$

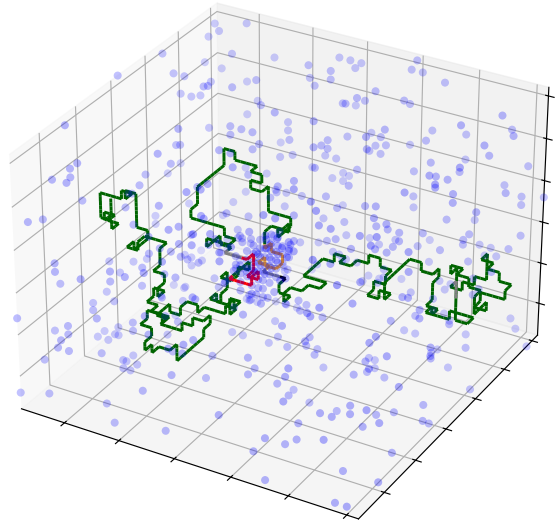
$\alpha = -1.5$

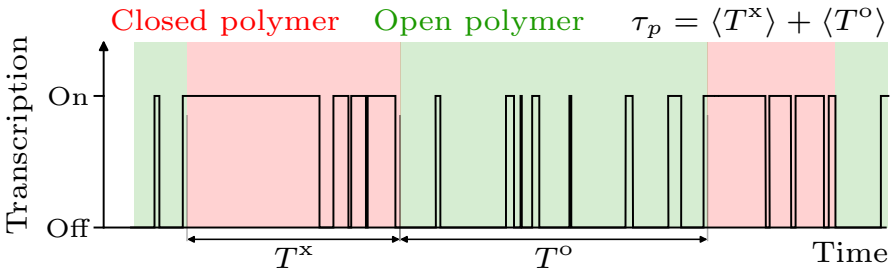
$\rho = -5.0$

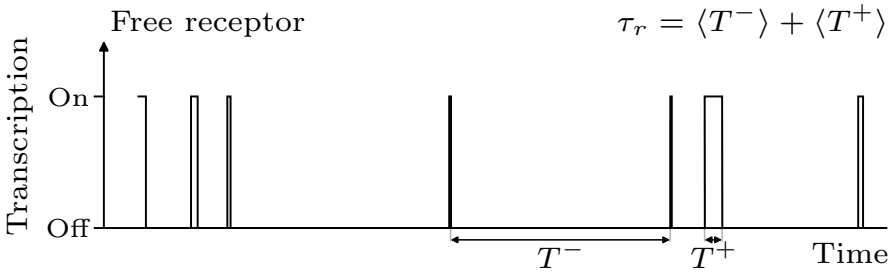
Transcription model

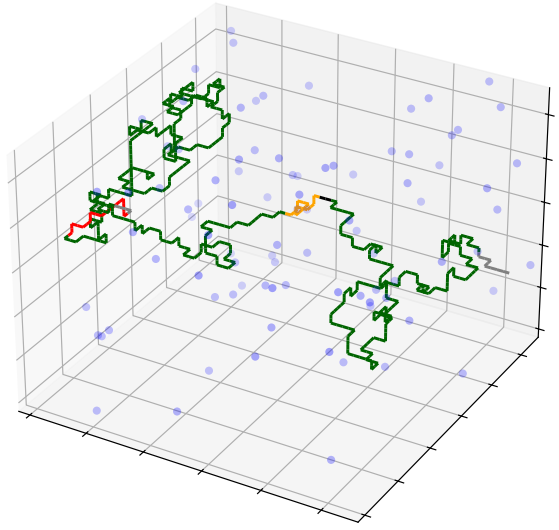


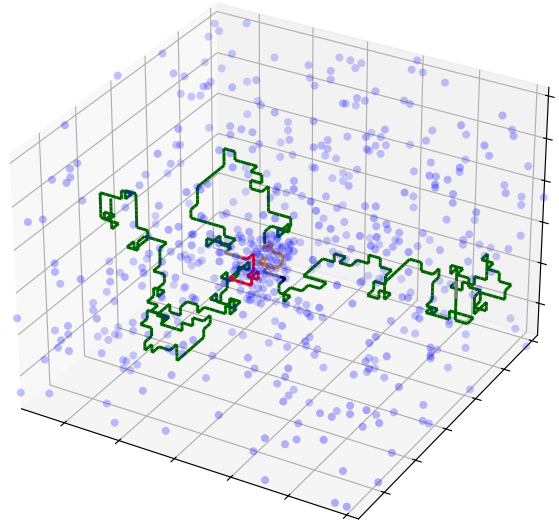


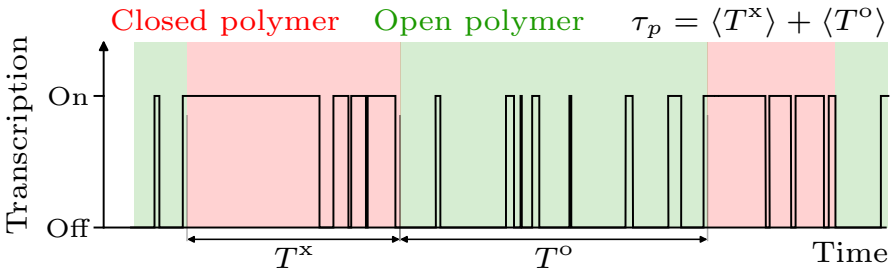


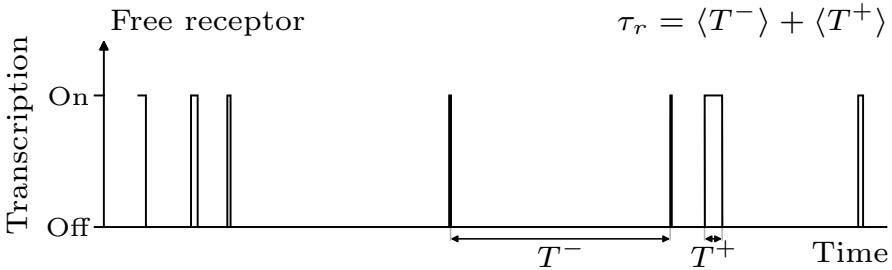

Closed polymer
Open polymer
$\tau_p = \langle T^{\mathrm{x}} \rangle + \langle T^{\mathrm{o}} \rangle$
Transcription
On
Off
$T^{\mathrm{x}}$
$T^{\mathrm{o}}$
Time

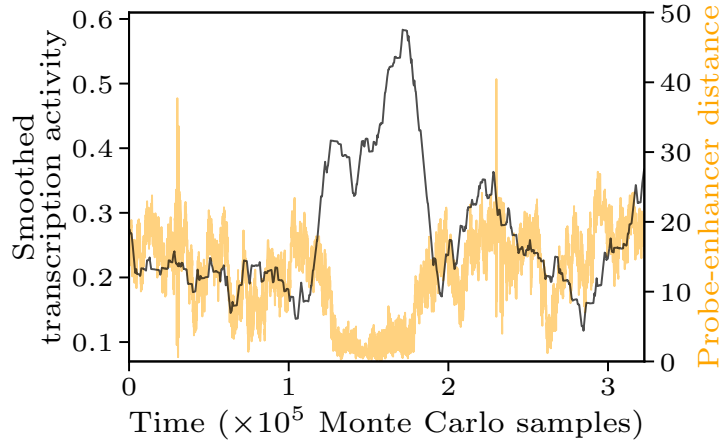

Free receptor
$\tau_r = \langle T^{-} \rangle + \langle T^{+} \rangle$
Transcription
On
Off
$T^{-}$
$T^{+}$
Time

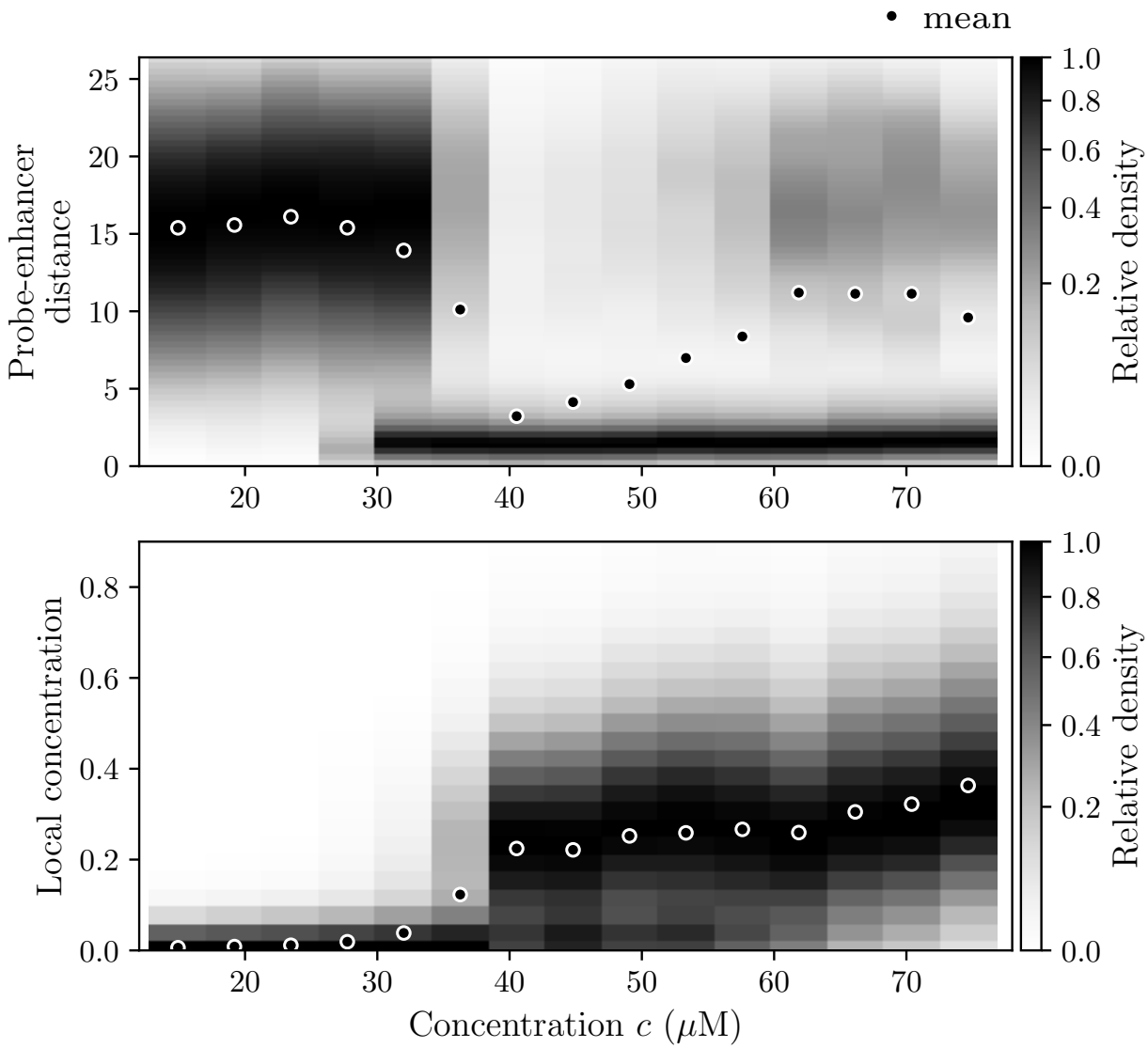


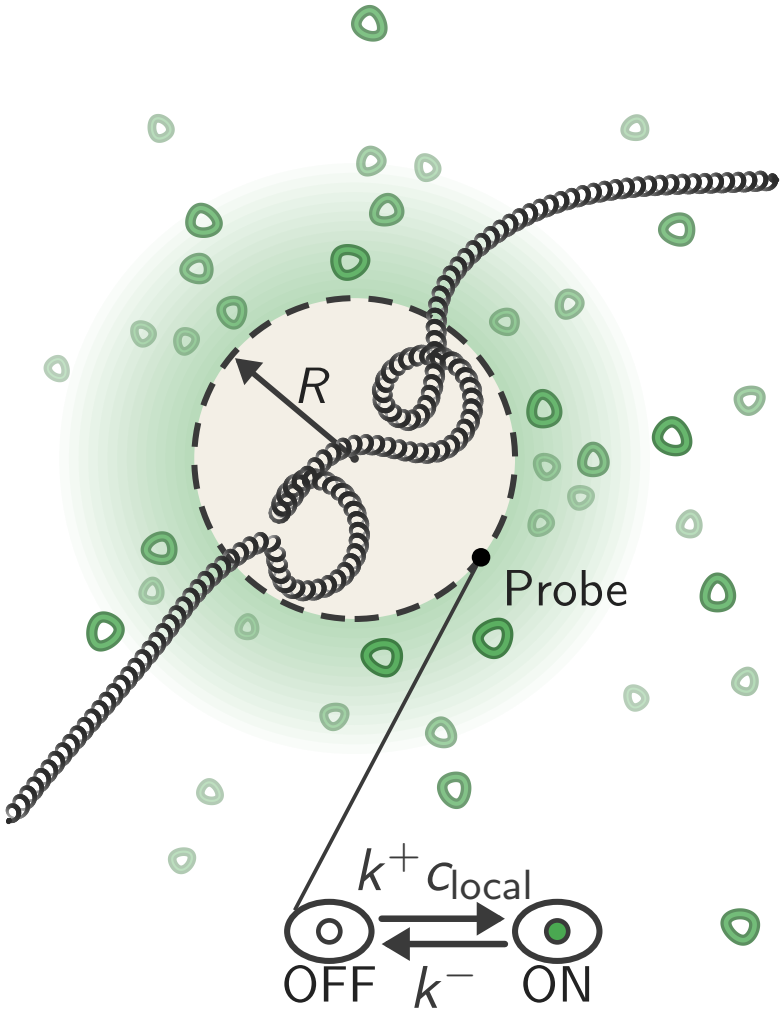



Closed polymer
Open polymer
$\tau_p = \langle T^{\mathrm{x}} \rangle + \langle T^{\mathrm{o}} \rangle$
Transcription
On
Off
$T^{\mathrm{x}}$
$T^{\mathrm{o}}$
Time


Free receptor
$\tau_r = \langle T^{-} \rangle + \langle T^{+} \rangle$
Transcription
On
Off
$T^{-}$
$T^{+}$
Time


Smoothed transcription activity
0.6
0.5
0.4
0.3
0.2
0.1
50
40
30
20
10
0
Probe-enhancer distance
0
1
2
3
Time ($\times 10^5$ Monte Carlo samples)


• mean
Probe-enhancer distance
Relative density
Local concentration
Concentration $c$ ($\mu$M)
distance




Flory–Huggins potential

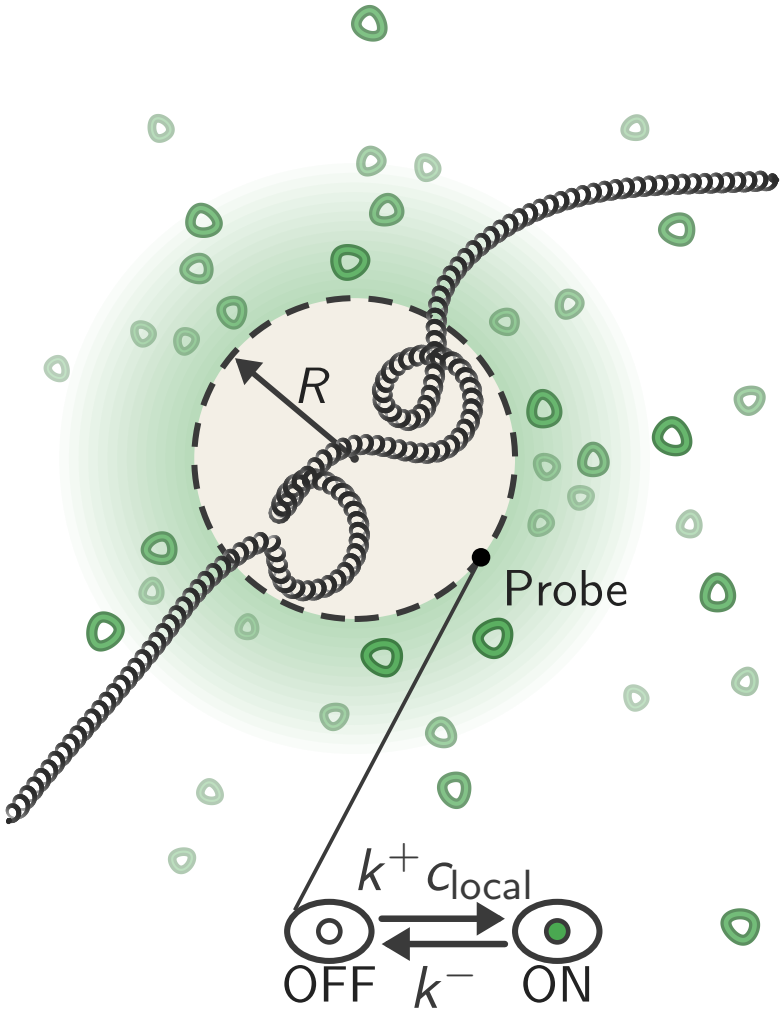
$$f(\phi) = \frac{\phi}{\nu} \ln \phi + (1-\phi) \ln(1-\phi) - \chi\phi^2 - \mu\phi$$

Cahn–Hilliard equation

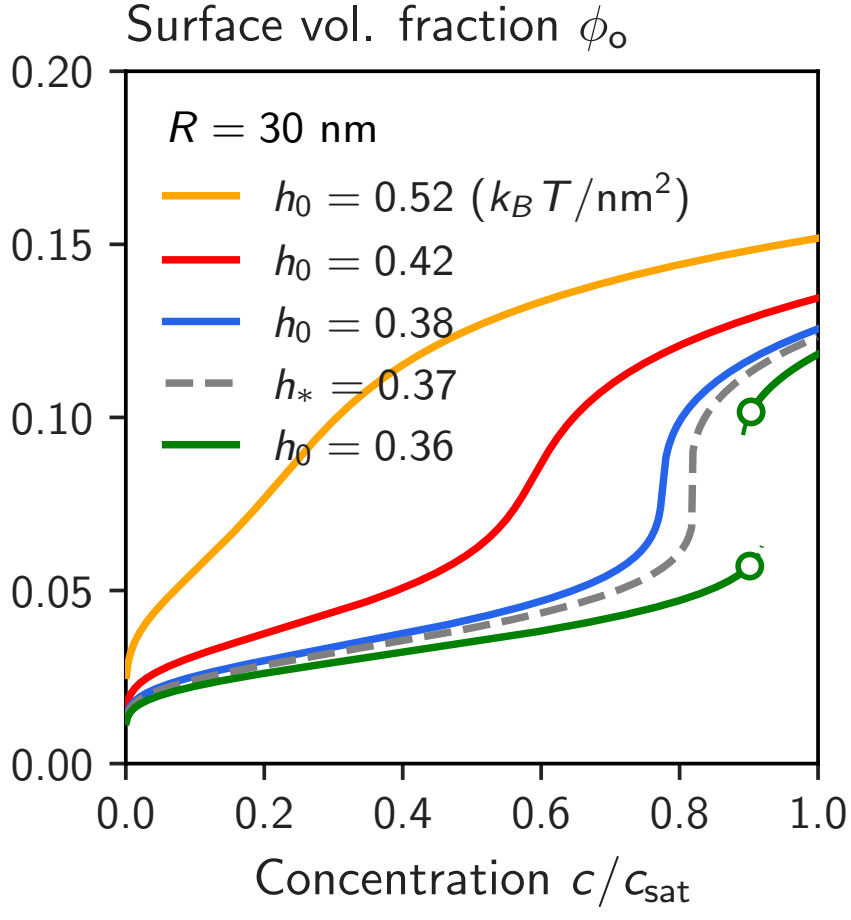
$$0 = \nabla^2 \left[ f'(\phi) - \kappa \nabla^2 \phi \right]$$

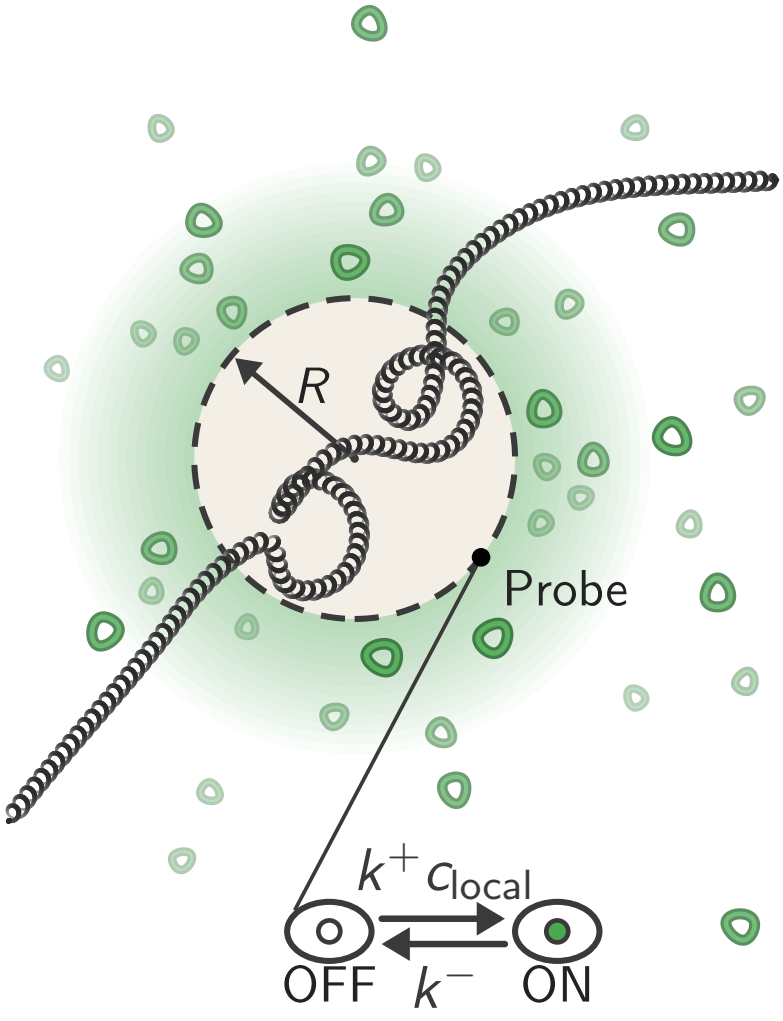
$$\kappa \, \partial_r \phi \, \big|_{r=R} = -h_0$$

| | |
|---|---|
| $\phi$ | TF volume fraction |
| $\nu$ | TF / solvent volume ratio |
| $\chi$ | TF–TF attraction |
| $\mu$ | chemical potential |
| $h_0$ | surface attraction |
| $\kappa$ | interface stiffness |

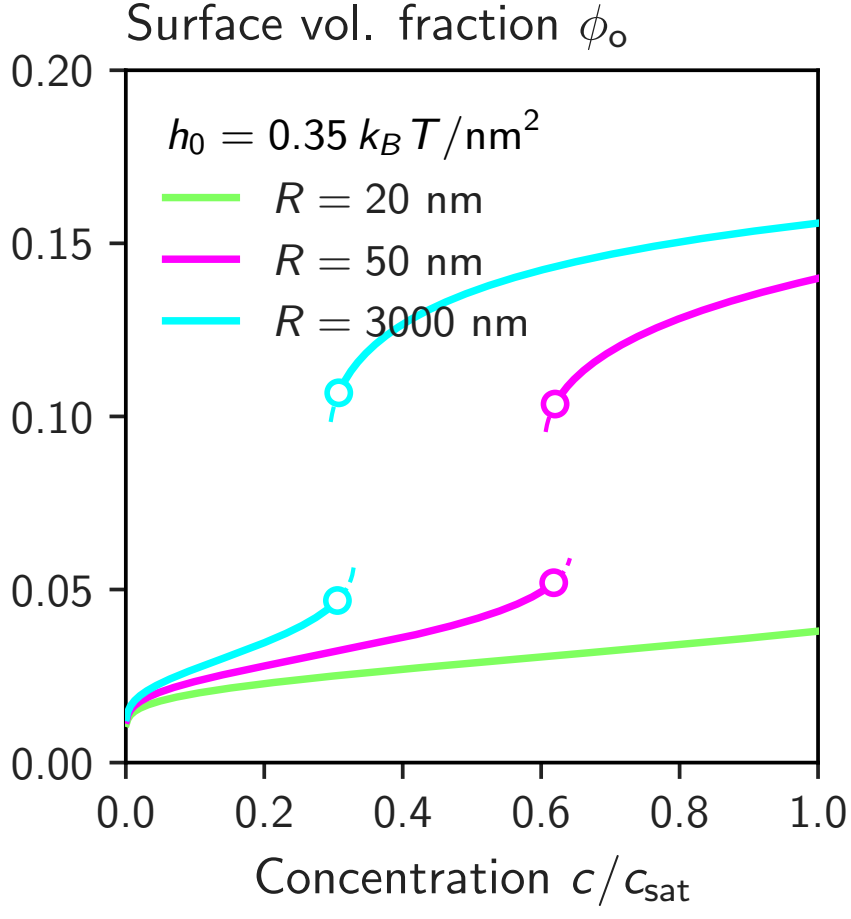

R
Probe
$k^+ c_{\text{local}}$
OFF
$k^-$
ON

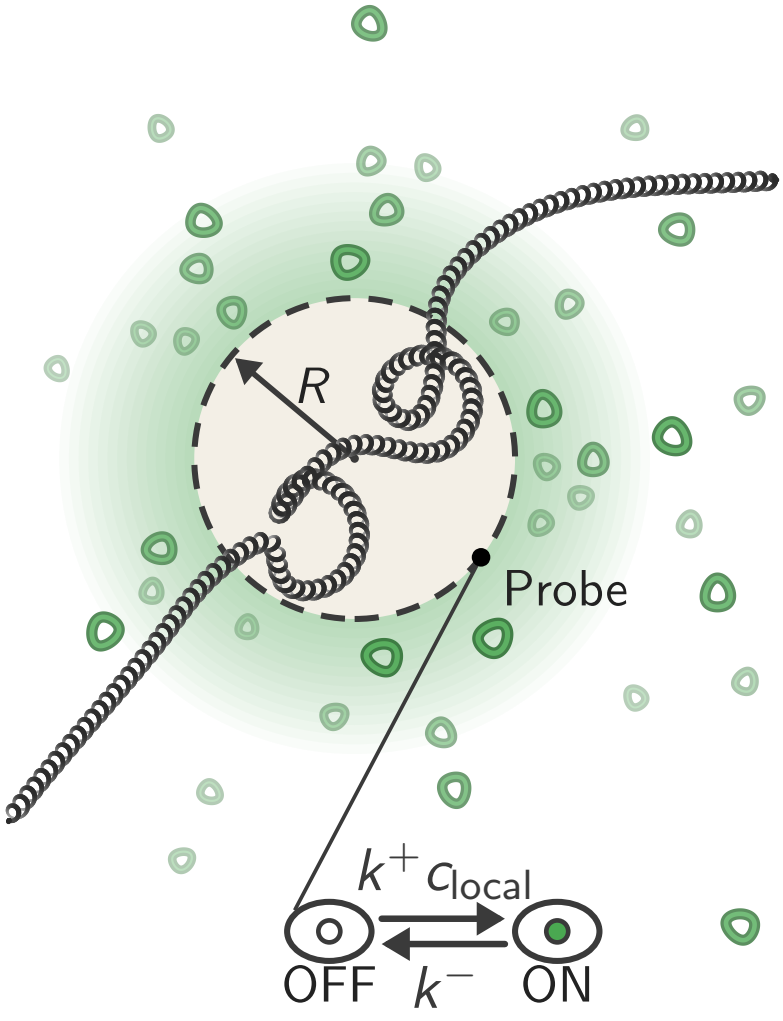
varying $h_0$


Surface vol. fraction $\phi_o$
0.20
0.15
0.10
0.05
0.00
$R = 30$ nm
$h_0 = 0.52$ ($k_B T/\text{nm}^2$)
$h_0 = 0.42$
$h_0 = 0.38$
$h_* = 0.37$
$h_0 = 0.36$
0.0
0.2
0.4
0.6
0.8
1.0
Concentration $c/c_{\text{sat}}$


R
Probe
$k^+ c_{\text{local}}$
OFF
$k^-$
ON

varying $R$


Surface vol. fraction $\phi_o$
0.20
0.15
0.10
0.05
0.00
$h_0 = 0.35\, k_B T/\text{nm}^2$
$R = 20$ nm
$R = 50$ nm
$R = 3000$ nm
0.0
0.2
0.4
0.6
0.8
1.0
Concentration $c/c_{\text{sat}}$



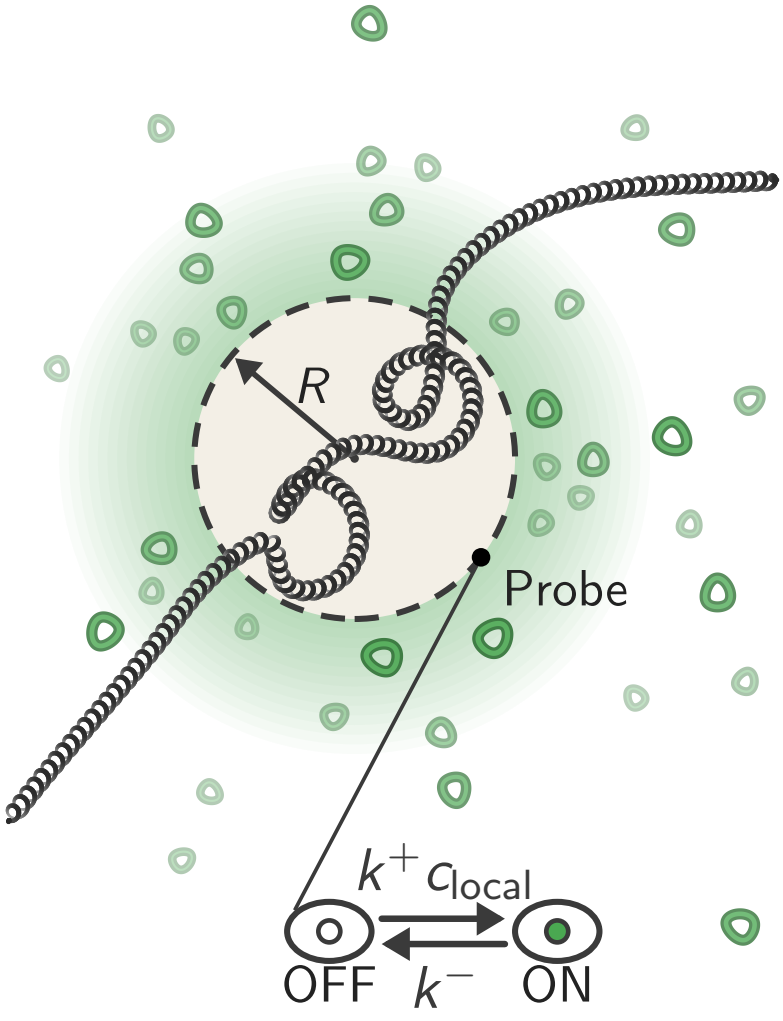

Free probe

$$\dot{\mathcal{F}}_{\text{free}}(c) = \frac{1}{\tau_r(c)}$$

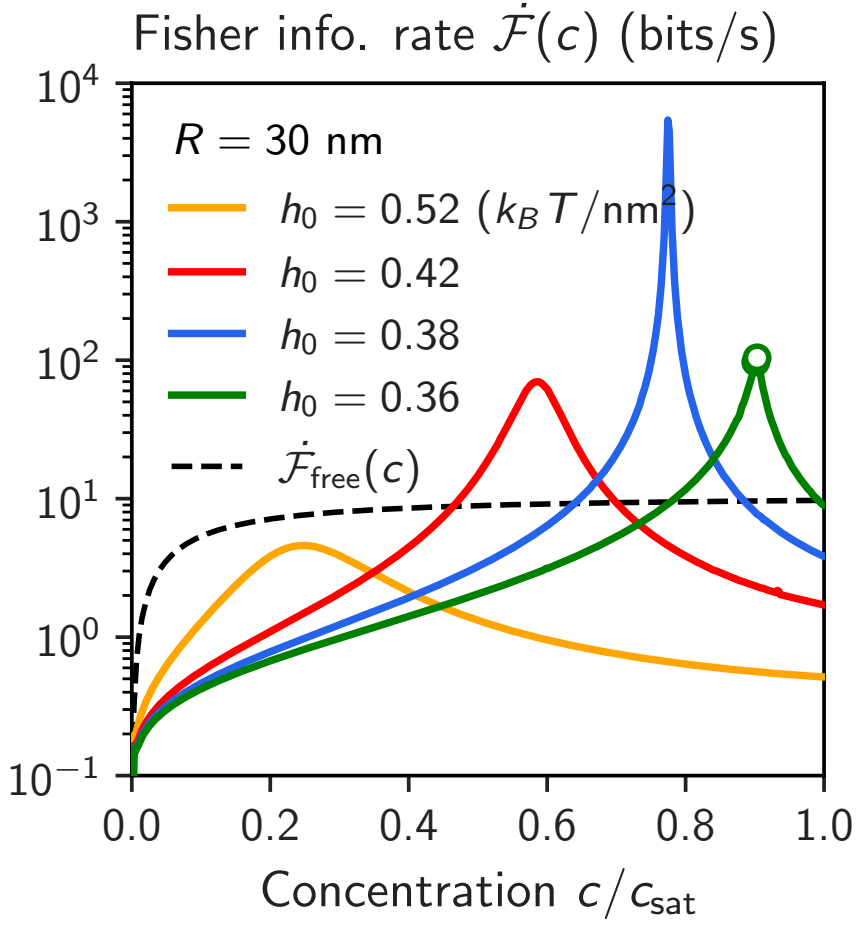
Probe on the anchor

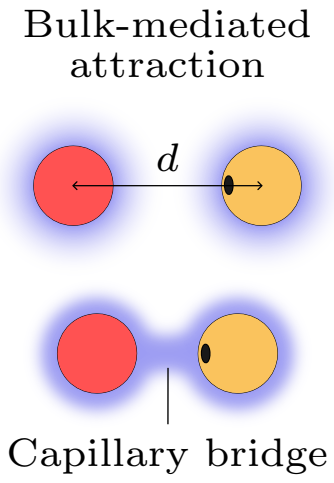
$$\dot{\mathcal{F}}(c) = \underbrace{\frac{1}{\tau_r(\phi)}}_{\text{probe}} \underbrace{\left(\frac{d \log \phi}{d \log c}\right)^2}_{\text{surface}}$$

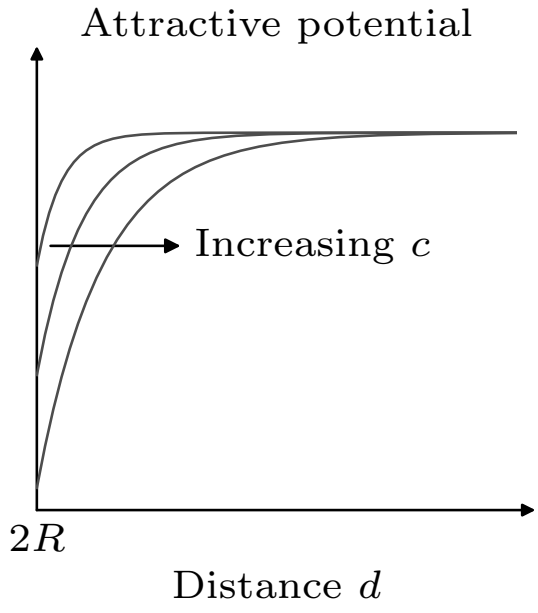
$\tau_r$ mean cycle time of the probe
$\phi$ TF volume fraction at the probe

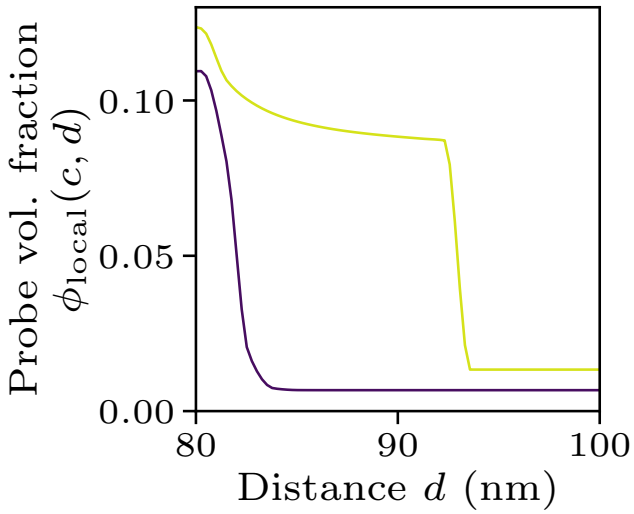

R
Probe
$k^+ c_{\text{local}}$
OFF
$k^-$
ON

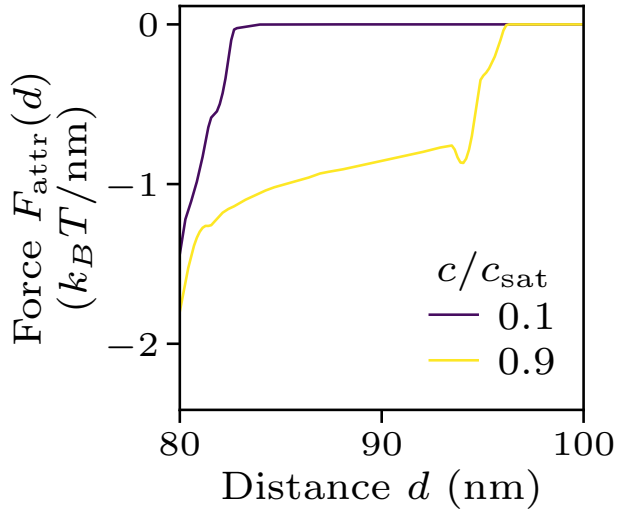
varying $h_0$

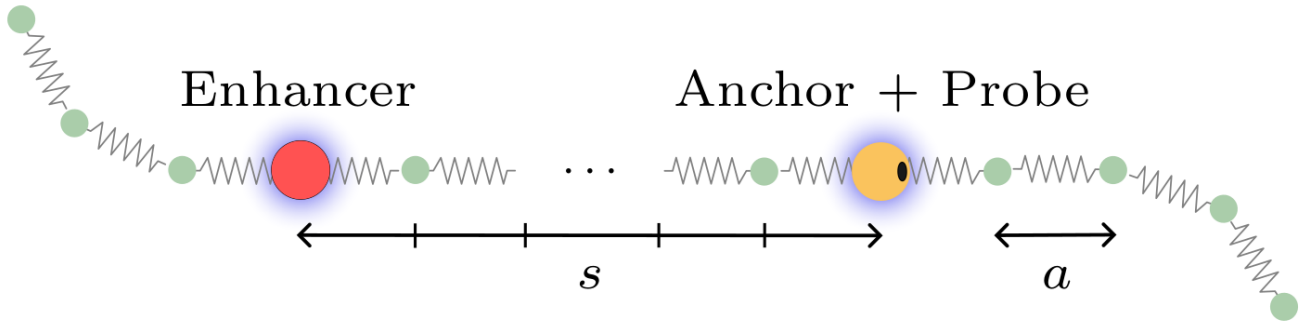

Fisher info. rate $\dot{\mathcal{F}}(c)$ (bits/s)
$10^4$
$10^3$
$10^2$
$10^1$
$10^0$
$10^{-1}$
$R = 30$ nm
$h_0 = 0.52$ ($k_B T$/nm$^2$)
$h_0 = 0.42$
$h_0 = 0.38$
$h_0 = 0.36$
$\dot{\mathcal{F}}_{\text{free}}(c)$
0.0
0.2
0.4
0.6
0.8
1.0
Concentration $c/c_{\text{sat}}$


Bulk-mediated attraction
d
Capillary bridge

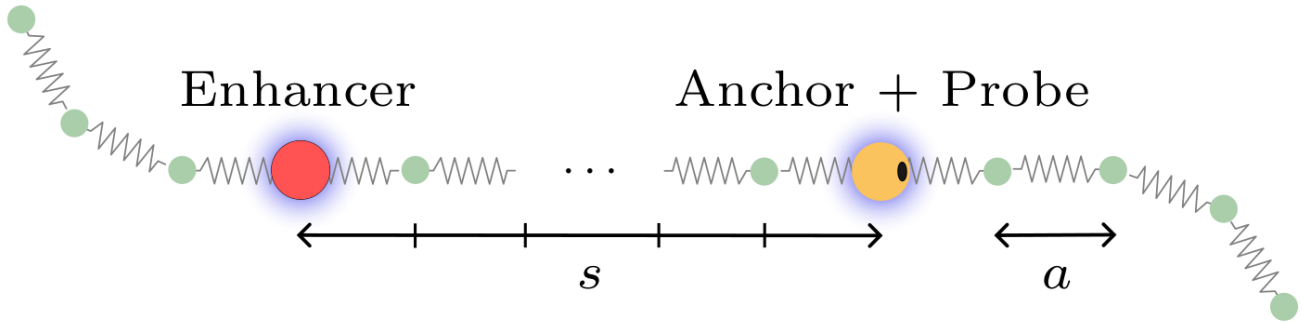

Attractive potential
Increasing c
2R
Distance d


Probe vol. fraction
$\phi_{\mathrm{local}}(c, d)$
0.10
0.05
0.00
80
90
100
Distance d (nm)


Force $F_{\mathrm{attr}}(d)$
$(k_B T/\mathrm{nm})$
0
−1
−2
$c/c_{\mathrm{sat}}$
0.1
0.9
80
90
100
Distance d (nm)

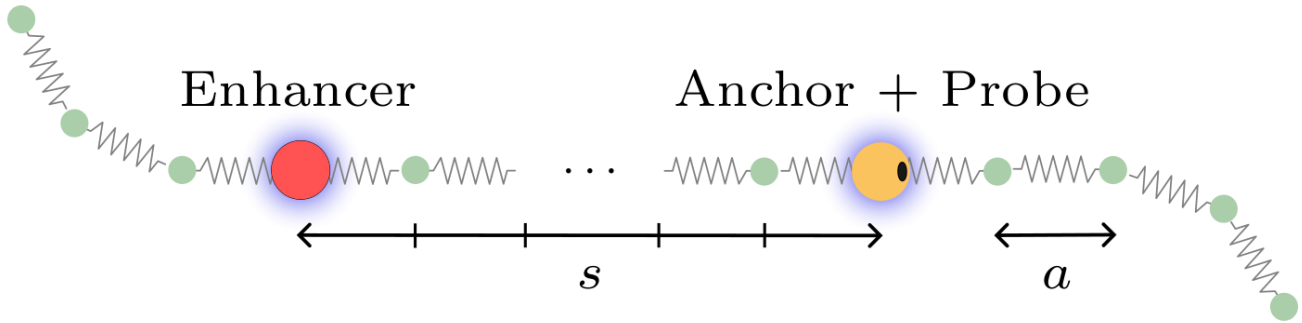



$$P_{\text{poly}}(d) \propto d^2 e^{-3d^2/2sa^2} \propto e^{-U_{\text{poly}}(d)/k_B T} \tag{1}$$




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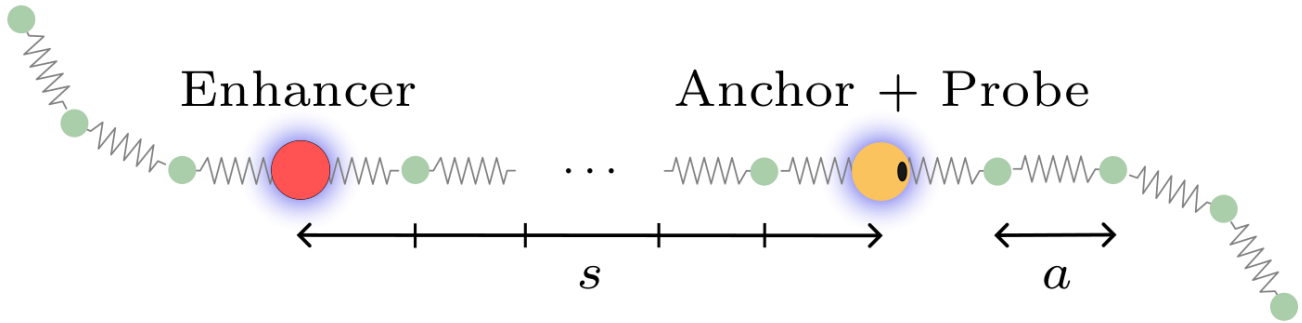
$$U_{\text{tot}}(d) = U_{\text{attr}}(d) + U_{\text{poly}}(d) \tag{2}$$




$$P_{\text{poly}}(d) \propto d^2 e^{-3d^2/2sa^2} \propto e^{-U_{\text{poly}}(d)/k_B T} \tag{1}$$

$$U_{\text{tot}}(d) = U_{\text{attr}}(d) + U_{\text{poly}}(d) \tag{2}$$

$$P_{\text{eq}}(d) \propto e^{-U_{\text{tot}}(d)/k_B T} \tag{3}$$

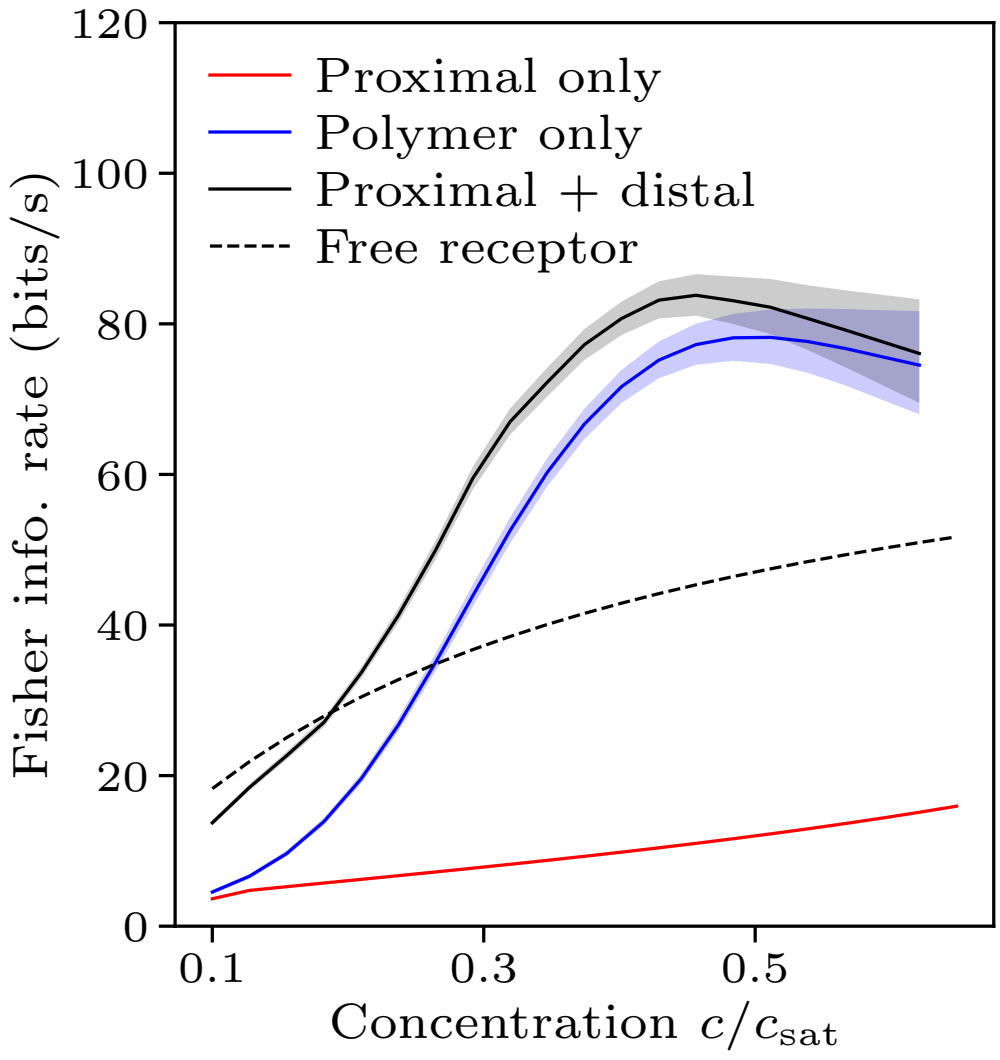



$$P_{\text{poly}}(d) \propto d^2 e^{-3d^2/2sa^2} \propto e^{-U_{\text{poly}}(d)/k_B T} \tag{1}$$

$$U_{\text{tot}}(d) = U_{\text{attr}}(d) + U_{\text{poly}}(d) \tag{2}$$

$$P_{\text{eq}}(d) \propto e^{-U_{\text{tot}}(d)/k_B T} \tag{3}$$

a: Kuhn bond length
s: genomic separation
d: euclidean distance


Proximal only
Polymer only
Proximal + distal
Free receptor
Fisher info. rate (bits/s)
120
100
80
60
40
20
0
0.1
0.3
0.5
Concentration $c/c_{\mathrm{sat}}$

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- Optimization forces us to think about constraints
- Different constraints can give rise to qualitatively different behavior
- Evolution need not optimize individual components: sensing is a distributed process
- Averages don't tell the whole story: single microscopic events can be extremely informative